

SS2 PHYSICS LESSON NOTES

FIRST TERM

1. Light Waves I
2. Light Waves I
3. Sound Waves
4. Human Eye
5. Midterm Examination
6. Application of Sound Waves
7. Midterm Break
8. Electric Charges
9. Description and Properties of Fields
10. Gravitational Field



WEEK 1 & 2 - LIGHT WAVES

INTRODUCTION TO LIGHT

Light is a form of **electromagnetic radiation** that can be detected by the human eye.

It exhibits **dual nature**:

- **Particle** (photon, as per quantum theory)
- **Wave** (as per classical and modern wave theory)

In this lesson, we focus on the **wave nature** of light.

NATURE OF LIGHT WAVES

Property	Description
Type	Electromagnetic wave
Nature	Transverse
Medium	Does not require a medium
Speed in Vacuum	3.0×10^8 m/s
Frequency Range	Around 4×10^{14} to 7.5×10^{14} Hz
Wavelength Range	Approximately 400 nm (violet) to 700 nm (red)

PROPERTIES OF LIGHT WAVES

1. Reflection of Light

- When light hits a smooth surface (like a mirror) and bounces back.
- **Laws of Reflection:**
 1. Angle of incidence = angle of reflection
 2. Incident ray, reflected ray, and normal lie in the same plane
- **Applications:** Mirrors, periscopes, rearview mirrors

2. Refraction of Light

- Bending of light when it passes from one transparent medium to another.
- Caused by a change in speed of light in different media
- **Laws of Refraction (Snell's Law):**

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

Where:

- i = angle of incidence
- r = angle of refraction
- v_1, v_2 = speeds in medium 1 and 2
- n = refractive index

Applications: Lenses, eyeglasses, camera lenses, prisms

Example 1:

A student in a lecture theatre can read from the board clearly but requires a pair of spectacles to read from a book.

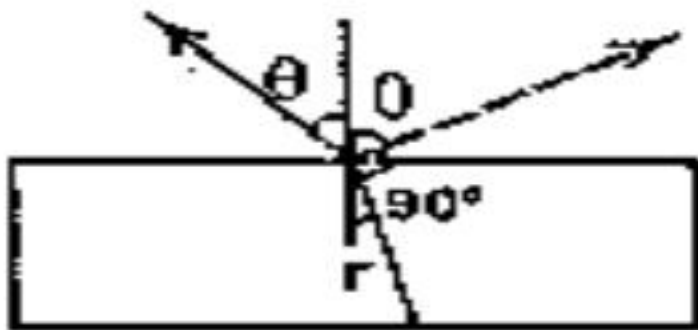
- What eye defect has this student?

Answer: Long sightedness or (Hypermetropia)/hyperopia

Example 2:

A ray of light is incident on air-glass boundary at an angle θ . If the angle between the partially reflected ray and the refracted ray is 90° , calculate θ , given that the refractive index of glass is 1.50.

Solution:



According to Shell's law

$$n = \sin\theta / \sin r$$

where $n = 1.5$ but $r + 90^\circ + \theta = 180^\circ$ (sum of angles on a straight line)

$$r = 180 - (90 + \theta)$$

$$r = 90^\circ - \theta$$

substitute $90^\circ - \theta$ for r in above

$$1.5 = \frac{\sin\theta}{\sin(90^\circ - \theta)}$$

$$\theta = 1.5(90^\circ - \theta)$$

$$\theta = \tan^{-1} [1.5]$$

$$= 56.31^\circ$$

3. Dispersion of Light

- Splitting of white light into its component colours when it passes through a prism.
- Occurs because different colours have different speeds in glass $\rightarrow$ different angles of refraction.
- **Colours in order:** Red, Orange, Yellow, Green, Blue, Indigo, Violet (**ROYGBIV**)

4. Diffraction of Light

- Bending or spreading of light waves around obstacles or through small openings.
- More noticeable with very small apertures and longer wavelengths.
- **Application:** Optical instruments, telescope design

5. Interference of Light

- Occurs when two light waves overlap to form a new wave pattern.
- **Types:**
 - **Constructive interference:** Bright fringes
 - **Destructive interference:** Dark fringes
- **Experiment:** Young's double-slit experiment
- **Application:** Anti-reflective coatings, holography

6. Polarization of Light

- Restriction of light wave oscillations to a single plane

- **Only transverse waves (like light) can be polarized**
- **Applications:** Polarized sunglasses, photography, 3D cinema glasses

Uses Of Polaroid

- It is used to eliminate reflected lights /rays
- It is useful in sachirimetry/measurement of concentration of sugar solution
- It is used in sunglasses to reduce the intensity of sunlight
- To eliminate reflected light
- Stress analysis/photo elasticity

CONCAVE AND CONVEX MIRRORS

- A **concave mirror** (also called a **converging mirror**) is a **spherical mirror** with its **reflecting surface curved inward**, like the interior of a spoon.
- A **convex mirror** (also called a **diverging mirror**) is a **spherical mirror** with its **reflecting surface curved outward**, like the back of a spoon.

PARTS OF A SPHERICAL MIRROR

- **Pole (P):** The center of the mirror's surface.
- **Center of Curvature (C):** The center of the sphere from which the mirror is made.
- **Radius of Curvature (R):** Distance between C and P.
- **Principal Axis:** A straight line passing through P and C.
- **Focus (F):** The point where parallel rays converge (concave) or appear to diverge from (convex) after reflection.
- **Focal Length (f):** Distance between P and F.

$$f = \frac{R}{2}$$

Concave Mirror – Image Formation for Different Object Positions

Object Position	Image Position	Nature of Image
At infinity	At focus (F)	Real, inverted, highly diminished
Beyond C	Between C and F	Real, inverted, diminished
At C	At C	Real, inverted, same size

Between C and F	Beyond C	Real, inverted, enlarged
At F	At infinity	Real, inverted, highly enlarged
Between F and P	Behind mirror	Virtual, erect, enlarged

Convex Mirror – Image Formation

- For all object positions, the image is:
 - Virtual
 - Erect
 - Diminished
 - Formed behind the mirror

Mirror Formula and Magnification

Mirror Formula:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Where: f = focal length, v = image distance, u = object distance
(sign conventions apply: real is positive, virtual is negative)

Magnification (M):

$$M = \frac{\text{Height of image}}{\text{Height of object}} = \frac{v}{u}$$

- ✓ For **concave mirror**, f is **positive** when image is real, and **negative** when virtual.
- ✓ For **convex mirror**, f is always **negative**.

Uses of Concave and Convex Mirrors

a. Concave Mirrors:

- In torches, flashlights, and car headlights (to produce a strong, parallel beam of light).
- As shaving/makeup mirrors (gives magnified upright image).
- In telescopes and solar cookers.

b. Convex Mirrors:

- As rear-view mirrors in vehicles (wide field of view).

- In security and surveillance (stores, hallways).
- At road junctions and blind corners.

Example 1:

An object is placed 30 cm in front of a concave mirror of focal length 10 cm. Find the position, nature, and magnification of the image.

Solution:

Given:

$u = -30$ cm, $f = -10$ cm (concave mirror uses sign convention)

Applying the formula, $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\frac{1}{-10} = \frac{1}{v} + \frac{1}{-30}$$

$$\frac{1}{-10} + \frac{1}{30} = \frac{1}{v}$$

$$\frac{-3+1}{30} = \frac{1}{v}$$

$$\frac{1}{v} = \frac{-2}{30} = \frac{-1}{15} \Rightarrow v = -15\text{cm}$$

Image is **real, inverted**, and **formed 15 cm** in front of the mirror.

$$M = \frac{v}{u} = \frac{-15}{-30} = 0.5$$

Image is **half the size** of the object.

Example 2:

The distance between a concave mirror and a object placed in front of it is 1.0m. If the radius of curvature of the mirror is 4.0m, the image formed will be

Solution

Recall: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

v = image distance, u = object distance; f = focal length

$$f = \frac{R}{2}$$

Given: $v = ?$ $u = 1.0\text{m}$

$$f = \frac{4}{2} = 2\text{m}$$

$$\frac{1}{v} = \frac{1}{2} - \frac{1}{1}$$

$$\frac{1}{v} = \frac{-1}{2}$$

$$v = -2$$

SPEED OF LIGHT IN DIFFERENT MEDIA

Medium	Approximate Speed (m/s)
Vacuum	3.0×10^8
Air	$\approx 2.98 \times 10^8$
Water	$\approx 2.25 \times 10^8$
Glass	$\approx 2.00 \times 10^8$

ELECTROMAGNETIC SPECTRUM

Light is part of the **electromagnetic spectrum**.

Other components (in order of increasing frequency):+

- Radio $\rightarrow$ Microwave $\rightarrow$ Infrared $\rightarrow$ **Visible Light** $\rightarrow$ Ultraviolet $\rightarrow$ X-ray $\rightarrow$ Gamma rays

APPLICATIONS OF LIGHT WAVES

Property	Application
Reflection	Mirrors, solar cookers
Refraction	Lenses, vision correction
Dispersion	Rainbow, spectrometry
Diffraction	Spectroscopy, telescopes
Interference	Fiber optics, sensors

Polarization	Sunglasses, LCD screens
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Factors that Affect Angle of Deviation

1. Refracting angle of Prism
2. Refracting index of the material of prism
3. Angle of Incidence

Example 1:

The angle of refraction(r) of a ray of white light from air through a triangular glass prism of refraction index 1.5 is 29.0° . Calculate the angle through which the ray is least deviated.

Solution

$$n = [\sin 1/2(A + D_m)] / [\sin 1/2(A)]$$

$$A = 2r = 2 \times 29 = 58$$

$$\sin 1/2(A + D_m) = n \sin 1/2(A)$$

$$1/2 (A + D_m) = \sin^{-1} (1.5 \times \sin 1/2 \times 58) = \sin^{-1} 0.7272 = 46.67$$

$$58 + D_m = 93.32$$

$$D_m = 35.32^\circ$$

Example 2:

Calculate the critical angle for the light travelling from glass to air.

[Refractive index of glass = 1.5]

Solution

$$\text{Refractive Index } (\eta) = \frac{1}{\sin C}$$

$$\eta = 1.5$$

$$1.5 = \frac{1}{\sin C}$$

$$\sin C = \frac{1}{1.5}$$

$$\sin C = 0.6667$$

$$C = \sin^{-1}(0.6667)$$

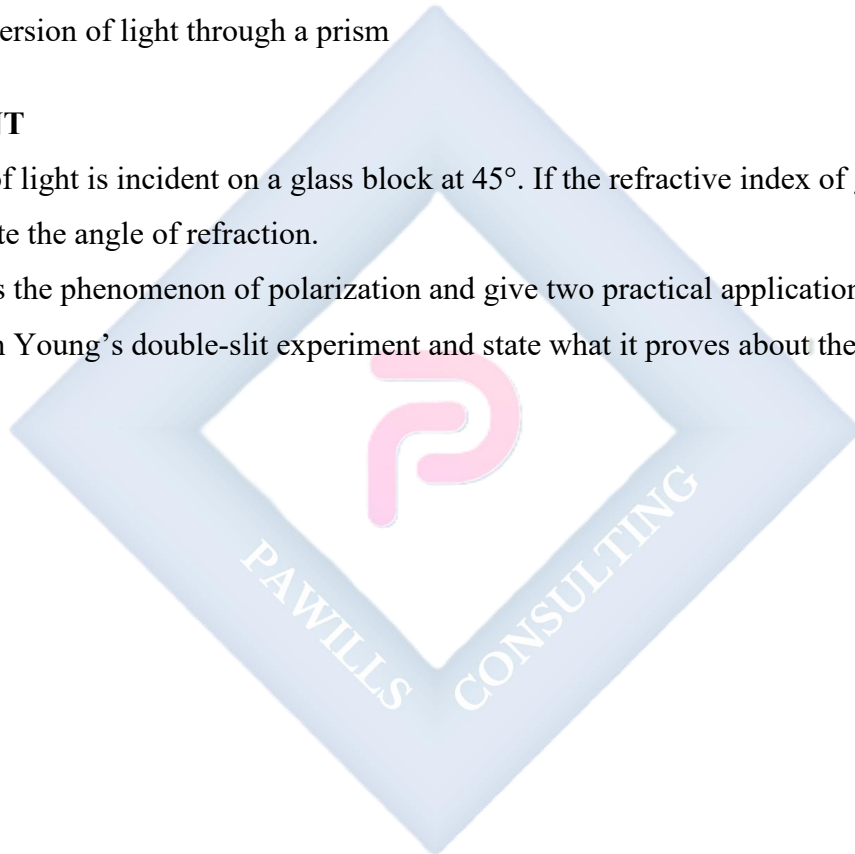
$$C = 41.81^\circ$$

EVALUATION

1. Define light as a wave and list three wave properties of light.
2. State and explain two laws of reflection.
3. Calculate the wavelength of light of frequency 5×10^{14} Hz
4. A light wave has a wavelength of 600 nm. Find its frequency.
5. Draw diagrams to show:
 - a) Reflection of light
 - b) Refraction of light+
 - c) Dispersion of light through a prism

ASSIGNMENT

1. A ray of light is incident on a glass block at 45° . If the refractive index of glass is 1.5, calculate the angle of refraction.
2. Discuss the phenomenon of polarization and give two practical applications.
3. Explain Young's double-slit experiment and state what it proves about the nature of light.



WEEK 3 - SOUND WAVES

MEANING OF SOUND AND SOUND WAVES

- **Sound** is a form of energy produced by vibrating objects and transmitted through a medium (solid, liquid, or gas) in the form of **longitudinal mechanical waves**.
- **Sound Waves** are longitudinal waves that propagate due to the compression and rarefaction of particles in a medium.

Characteristics:

- They require a **material medium** to propagate (i.e., cannot travel in a vacuum).
- They are **mechanical** and **longitudinal** in nature.
- Sound waves move faster in solids than in liquids and gases.

PRODUCTION OF SOUND

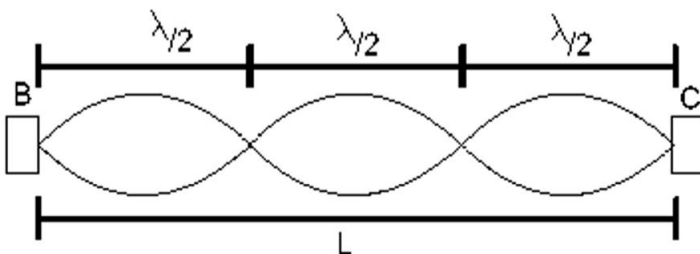
- Sound is produced by **vibrating bodies**, such as vocal cords, strings of musical instruments, and loudspeakers.
- The vibration causes the surrounding air particles to vibrate, producing compressions and rarefactions.

PROPAGATION OF SOUND

Sound travels by the successive compressions and rarefactions of particles in a medium.

- **Compression:** Region of high pressure where particles are close together.
- **Rarefaction:** Region of low pressure where particles are far apart.
- **Resonance:** This is the effect felt/observed when a body vibrating with its own natural frequency transfers its frequency to the bodies around it, such that they all vibrate with the same frequency.

SOUND VIBRATION IN STRINGS



Second overtone

$$1 = \frac{3\lambda}{2}$$
$$F_3 = 3F_0$$

CHARACTERISTICS OF SOUND WAVES

1. **Frequency (f):** Number of vibrations per second (Hz).
2. **Amplitude (A):** Maximum displacement of the wave from equilibrium – determines loudness.
3. **Wavelength (λ):** Distance between two successive compressions or rarefactions.
4. **Speed (v):** Rate at which sound travels through a medium.

$$V = f\lambda$$

RANGE OF HEARING

- **Infrasonic:** Below 20 Hz (e.g., earthquakes).
- **Audible:** 20 Hz – 20,000 Hz (Human range).
- **Ultrasonic:** Above 20,000 Hz (e.g., sonar, medical ultrasound).

SPEED OF SOUND IN DIFFERENT MEDIA

Medium	Speed (m/s)
Air (at 20°C)	343
Water	1,480
Iron	5,120

Note: Sound travels fastest in solids and slowest in gases because of particle density and elasticity.

BEHAVIOUR OF SOUND WAVES

1. **Reflection of Sound:**
 - Produces **echoes** when sound reflects off a hard surface.
 - Echo time gap must be ≥ 0.1 sec to be heard separately.
2. **Refraction of Sound:**

- Occurs when sound passes through different layers (e.g., warm to cold air).
- Changes direction due to change in speed.

3. **Diffraction of Sound:**

- Bending of sound waves around obstacles or through openings.
- Explains why we can hear sounds around corners.

4. **Interference of Sound:**

- Occurs when two waves superimpose to form a resultant wave.
- Leads to loud (constructive) or soft (destructive) sounds.

APPLICATIONS OF SOUND WAVES

1. **Sonar (Sound Navigation and Ranging):**

- Used in ships/submarines to detect underwater objects using **ultrasonic waves**.

2. **Ultrasound in Medicine:**

- Imaging of internal organs (e.g., foetus in the womb).

3. **Industrial Non-Destructive Testing:**

- Detecting cracks or flaws in materials.

4. **Sound Communication:**

- Speech, telecommunication, public address systems.

SAMPLE CALCULATIONS

Example 1:

A sound wave has a frequency of 250 Hz and a wavelength of 1.4 m. Calculate its speed.

$$V = f\lambda = 250 \times 1.4 = 350 \text{ m/s}$$

Example 2:

If the speed of sound in air is 340 m/s and the wavelength is 0.85 m, find the frequency.

$$F = v\lambda = 340 \times 0.85 = 400 \text{ Hz}$$

Example 3:

A sonometer wire has a frequency of 259Hz. It is sounded alongside a tuning fork of frequency 256Hz. Calculate the beat frequency.

Solution

$$\text{Beat frequency} = f_1 - f_2 = (259 - 256)\text{Hz} = 3\text{Hz}$$

EVALUATION

1. Define sound and explain how it is produced.
2. Mention three characteristics of sound waves.
3. Differentiate between ultrasonic and infrasonic sounds.
4. What is the speed of sound in air, water, and iron?
5. Describe how echoes are formed.
6. Calculate the frequency of a sound wave with a speed of 330 m/s and wavelength of 1.1m.
7. Explain one application of ultrasonic waves in medicine.

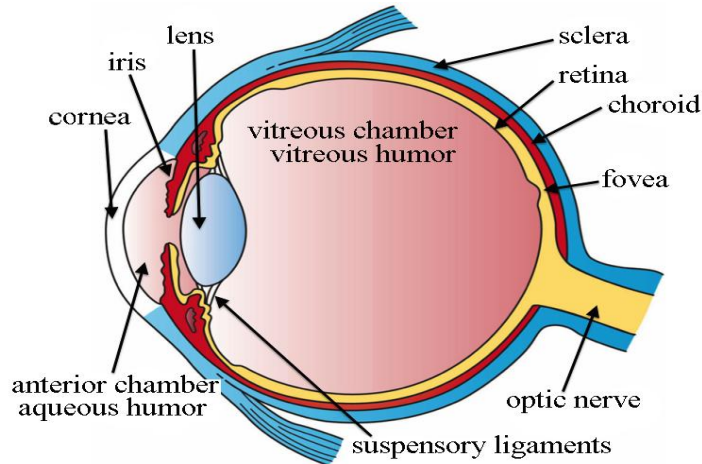
ASSIGNMENT

1. List five differences between sound waves and light waves.
2. Draw a labeled diagram of a longitudinal wave showing compression and rarefaction.
3. If an echo is heard 1.5 seconds after a sound is made, and the speed of sound is 340 m/s, find the distance to the wall.

WEEK 4 - HUMAN EYE

STRUCTURE OF THE HUMAN EYE

The human eye is a **natural optical instrument** that enables us to see by detecting light and converting it into electrical signals sent to the brain.



Major Parts of the Human Eye and Their Functions:

Part	Function
Cornea	Transparent front layer that helps focus light into the eye
Iris	Coloured part that controls the size of the pupil (thus controlling light entry)
Pupil	Opening in the centre of the iris through which light enters
Lens	Transparent, flexible, convex lens that focuses light on the retina
Ciliary muscles	Alter the shape (curvature) of the lens to focus objects (accommodation)
Retina	Light-sensitive layer that contains photoreceptor cells (rods and cones); it converts light into electrical impulses
Optic nerve	Carries the electrical impulses from the retina to the brain
Aqueous humor	Fluid between the cornea and lens; helps maintain pressure and nourishes the eye
Vitreous humor	Gel-like substance between the lens and retina; helps maintain the shape of the eye
Sclera	White, protective outer layer of the eye

IMAGE FORMATION IN THE HUMAN EYE

- Light from an object enters the eye through the **cornea**, passes through the **aqueous humor**, **pupil**, **lens**, and **vitreous humor**, then falls on the **retina**.
- The **lens** converges light rays to form a **real, inverted, and diminished image** on the retina.
- The **retina** converts the image into electrical signals, which are carried to the brain via the **optic nerve**.
- The **brain** interprets the image as **upright and properly oriented**.

ACCOMMODATION AND DEFECTS OF VISION AND THEIR CORRECTIONS

Accommodation is the **ability of the eye to adjust the focal length of the lens** to focus on near or distant objects.

- For **near objects**, the **ciliary muscles contract**, making the lens **thicker and more convex**.
- For **distant objects**, the **ciliary muscles relax**, making the lens **flatter**.

Defect	Description	Cause	Correction
Short-sightedness (Myopia)	Can see near objects clearly, but distant ones appear blurred	Eyeball too long or lens too curved	Use of concave lens to diverge light rays
Long-sightedness (Hypermetropia)	Can see distant objects clearly, but near ones appear blurred	Eyeball too short or lens too flat	Use of convex lens to converge light rays
Astigmatism	Blurred or distorted vision at all distances	Irregular curvature of the cornea or lens	Use of cylindrical lenses
Presbyopia	Inability to focus on near objects due to aging	Weakening of ciliary muscles and lens rigidity	Use of bifocal lenses

Cataract	Clouding of the lens	Aging or injury	Surgery (lens replacement)
Color blindness	Inability to distinguish certain colors	Genetic (defect in cone cells)	No full correction; use of color filters or special glasses

SIMILARITIES AND DIFFERENCES BETWEEN THE HUMAN EYE AND A CAMERA

Similarities

Feature	Human Eye	Camera
Image forming part	Lens	Camera lens
Image-sensitive surface	Retina	Film or digital sensor
Light control	Iris and pupil	Diaphragm and aperture
Focusing	Ciliary muscles change lens shape	Camera lens moves
Protection	Eyelids and tear glands	Camera casing and lens cover

Differences:

- The eye uses a flexible lens to focus; the camera uses a movable lens.
- The retina in the eye acts as the screen; the camera uses a photographic film or sensor.

CARE OF THE EYES

1. Avoid reading in dim light.
2. Do not look directly at bright light (e.g., sun, lasers).
3. Visit an optometrist regularly.
4. Eat vitamin A-rich foods (e.g., carrots, spinach).
5. Avoid prolonged screen time.
6. Wear protective glasses when necessary.

Evaluation Questions

1. List five parts of the human eye and state their functions.
2. Explain how the eye forms an image.
3. What is accommodation?
4. Describe two defects of vision and their corrections.
5. 4. NECO June 2012 (Paper 1, Q20)

The part of the eye that controls the amount of light entering it is the:

- A. retina
- B. cornea
- C. iris
- D. lens

6. 7. WAEC June 2011 (Paper 1, Q19)

Which of the following defects of the eye is caused by the eyeball being too long?

- A. Astigmatism
- B. Long-sightedness
- C. Short-sightedness
- D. Colour blindness

Assignment

1. WAEC June 2017 (Paper 2, Q5a)

Name five parts of the human eye and state the function of each.

2. WAEC June 2009 (Paper 2, Q3a)

Explain how the eye accommodates to focus on near and distant objects.

3. WAEC June 2013 (Paper 2, Q6a)

State the defect of vision corrected by a concave lens and how it works.

4. NECO June 2019 (Paper 2, Q4a)

What type of lens is used to correct long-sightedness? Explain how it works.

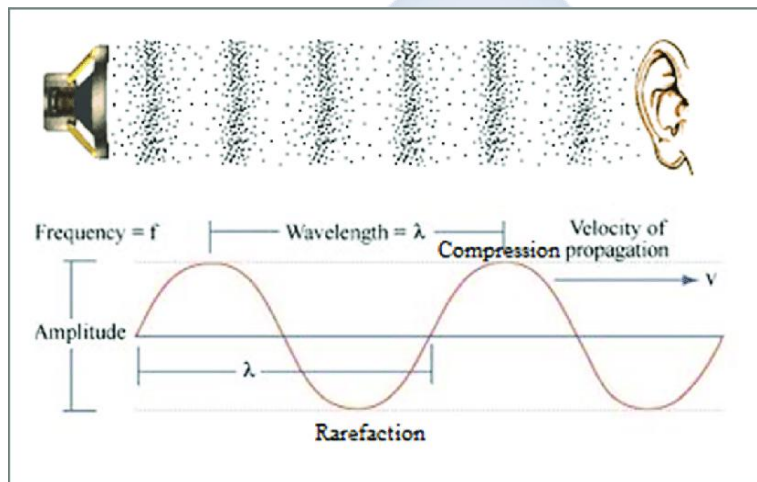
5. NECO June 2015 (Paper 2, Q6b)

Describe briefly how a person suffering from astigmatism can have their vision corrected.

WEEK 6 - APPLICATION OF SOUND WAVE

SCIENTIFIC PROPERTIES THAT ENABLE APPLICATION OF SOUND WAVES

- **Reflection of sound** (echoes) – used in SONAR and echolocation
- **Refraction** – used in sound focusing (medical and industrial applications)
- **Diffraction** – sound bending around corners (important in speaker design)
- **Interference** – used in noise-cancelling headphones
- **Frequency and wavelength** – ultrasonic vs infrasonic waves
- **Amplitude and intensity** – loudness detection, used in alarms and meters.



FIELDS OF APPLICATION

1. Medicine

a. Diagnostic Imaging – Ultrasound (Sonography):

- High-frequency sound waves (above 20 kHz) are directed into the body.
- Echoes from tissues of varying densities are recorded to form an image.
- Common for prenatal imaging, liver, kidney, and heart examinations.
- **Advantages:** Safe (non-ionizing), real-time, painless
- **Limitation:** Cannot penetrate bones or gas-filled cavities

b. Therapeutic Ultrasound:

- Used in physiotherapy to treat muscle injuries, reduce pain, and promote healing
- Vibrations generate localized heat in tissues
- Also used for breaking up kidney stones (lithotripsy)

2. Communication

a. Microphones and Loudspeakers:

- Microphones convert sound (mechanical wave) to electrical signals
- Speakers do the reverse, converting electrical signals to audible sound
- Used in: Phones, PA systems, radios, hearing aids

b. Sound in Telecommunication:

- Modulation of sound waves for long-distance transmission
- Sound wave conversion used in GSM and voice recognition systems

c. Voice-controlled Systems:

- Smart assistants like **Siri, Alexa, Google Assistant** use sound wave input
- Commands are processed using sound recognition algorithms

3. Industry and Engineering

a. Ultrasonic Cleaning:

- High-frequency vibrations agitate cleaning solution
- Loosens dirt from delicate items like jewelry, watches, circuit boards

b. Non-Destructive Testing (NDT):

- High-frequency waves detect internal flaws/cracks in materials
- Applied in aircraft maintenance, oil pipelines, bridges, buildings

c. Sound-based Welding:

- Ultrasonic welding used to join plastics and even metals in precise applications

4. Music and Entertainment

a. Musical Instruments:

- Vibrations from strings, membranes, or air columns produce musical sounds
- String (violin), wind (flute), and percussion (drum) instruments

b. Audio Technology:

- Speakers, earphones, and recording studios use sound wave manipulation
- Equalizers modify amplitude and frequency for better audio quality

5. Security and Surveillance

a. Ultrasonic Motion Detectors:

- Emit high-frequency sound waves
- Detect motion by sensing changes in the reflected sound pattern

b. Burglar Alarms and Sound Sensors:

- Alarm systems detect specific sound frequencies (e.g., glass shattering)
- Used in banks, homes, and safes

6. Nature and Biology

a. Echolocation:

- Used by bats, dolphins, whales, and some birds
- Animals emit ultrasonic sounds and analyze returning echoes to:
 - Avoid obstacles
 - Locate prey
 - Navigate in darkness

b. Animal Communication:

- Elephants use infrasonic waves to communicate over long distances
- Some animals detect natural sound signals like thunder or predator movements

7. Seismology and Earthquake Monitoring

- **Seismic waves** (a form of sound waves) used to study Earth's interior
- Geophones detect low-frequency sound waves from earthquakes
- Help in:
 - Earthquake early warning systems
 - Oil and mineral exploration
 - Volcanic monitoring

NAVIGATION, DETECTION & RANGING

a. SONAR (Sound Navigation and Ranging): This is a pulse-wave techniques that uses sound wave to navigate ships and submarines and detects depth of water.

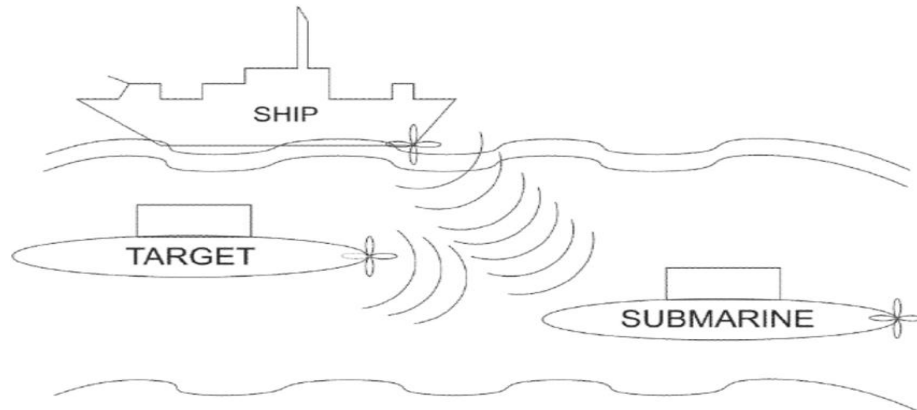
- Emits sound pulses underwater and detects echoes to determine object position.
- Used in:
 - Submarines for underwater navigation
 - Fishing vessels to locate schools of fish
 - Ocean floor mapping
- **Formula:**

$$\text{Distance} = \frac{\text{Velocity of Sound} \times \text{Time}}{2}$$

(Divided by 2 because sound travels to the object and back)

b. Radar vs SONAR:

- Radar uses radio waves (electromagnetic), SONAR uses sound (mechanical).
- SONAR is preferred underwater due to better propagation of sound in water.



LIMITATIONS OF SOUND WAVE APPLICATIONS

- Cannot travel through a vacuum (requires medium)
- Affected by obstacles, temperature, pressure, and humidity
- High-frequency ultrasonic equipment is costly
- Noise pollution and distortion in communication
- Limited range compared to electromagnetic waves (radio, light)

EVALUATION

1. Define SONAR and state one use in industry.
2. How does ultrasound work in medical imaging?
3. Explain how echolocation works and list two animals that use it.
4. Mention three limitations of sound wave technology.
5. Distinguish between ultrasonic and infrasonic sound with examples.

ASSIGNMENT

Research Report (1 page):

Investigate how sound waves are used in **oil and gas exploration**. Include:

- Tools used (e.g., geophones, seismic survey trucks)

- How reflected seismic waves help find underground resources
- Environmental concerns



WEEK EIGHT - ELECTRIC CHARGES

DEFINITION AND NATURE OF ELECTRIC CHARGE

Electric Charge is a fundamental property of matter that causes it to experience a force when placed in an electromagnetic field. It is responsible for electric and magnetic phenomena.

Microscopic View:

- Atoms consist of **protons** (+), **electrons** (−), and **neutrons** (neutral).
- **Protons** and **electrons** carry equal and opposite charges.
- The **net charge** of an atom is normally zero (neutral).
- When electrons are gained or lost, the body becomes **charged**. SI Unit: Coulomb (C)
- Symbol: q or Q

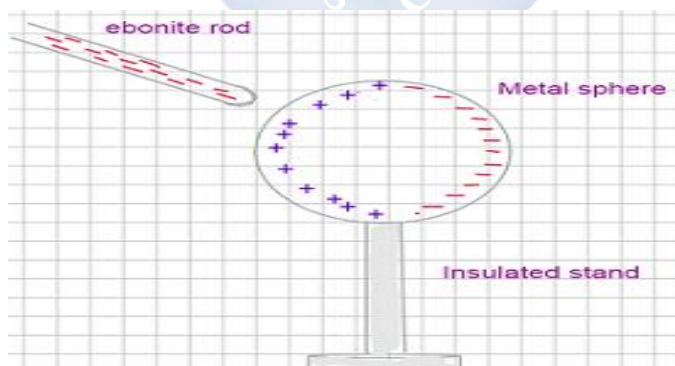
TYPES OF ELECTRIC CHARGES

1. **Positive Charges (+):** Due to the **loss of electrons**. Found in **protons**.
2. **Negative Charges (−):** Due to the **gain of electrons**. Found in **electrons**

BASIC LAWS OF ELECTRIC CHARGES

1. **Like charges repel** each other.
2. **Unlike charges attract** each other.

Law of Charges: Like charges repel, unlike charges attract.



METHODS OF CHARGING A BODY

1. Charging by Friction

- When two different materials are rubbed together, one loses electrons, the other gains.

- **Examples:**

- Glass rod + Silk → glass becomes positively charged.
- Ebonite rod + Fur → ebonite becomes negatively charged.

2. Charging by Conduction (Contact)

- A charged object touches a neutral object.
- Charges flow from the charged to the neutral body.
- The neutral body becomes **similarly charged**.

Charging by Induction

- A charged object is brought close to a neutral object (without touching).
- Charges in the neutral object redistribute.

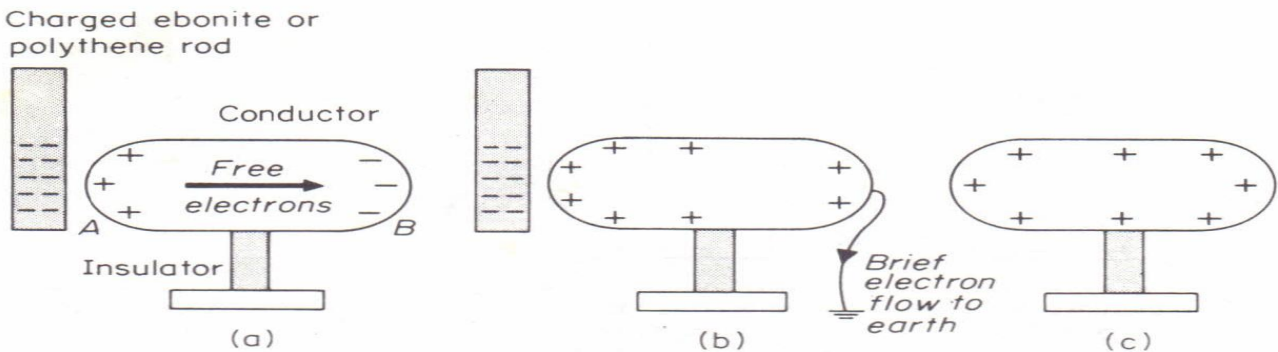


Fig. 32.5. Charging positively by induction

- The object is **earthed** (grounded), allowing electrons to move.
- On removing the original charged body, the neutral object becomes oppositely charged.

PROPERTIES OF ELECTRIC CHARGES

1. **Two types only:** Positive and negative.
2. **Additivity:** Charges add algebraically (e.g., $+5\text{ C} + -3\text{ C} = +2\text{ C}$).
3. **Conservation of Charge:** Charges can neither be created nor destroyed, only transferred.
4. **Quantization of Charge:**
 - Charge exists in discrete units.
 - Smallest unit is **elementary charge (e)**:

$$e = 1.6 \times 10^{-19}\text{ C}$$

$$q = ne, \text{ where } n \text{ is an integer}$$

Example:

$$q = 5 \times 1.6 \times 10^{-19} = 8.0 \times 10^{-19} \text{ C}$$

5. **Forces between charges:** Charges exert attractive or repulsive **electrostatic forces** governed by **Coulomb's Law**.

LAW OF CONSERVATION OF CHARGE

The total electric charge in an isolated system remains constant.

- In chemical reactions, decay, or charge transfer, the total charge before and after remains unchanged.

CONDUCTORS, SEMICONDUCTORS, AND INSULATORS

Type	Definition	Examples
Conductors	Materials that allow free movement of charges	Copper, Aluminium, Graphite
Insulators	Materials that do not allow charge movement	Rubber, Glass, Plastic, Wood
Semiconductors	Conduct poorly at low temperature, conduct better when heated or doped	Silicon, Germanium

DETECTION OF ELECTRIC CHARGE – GOLD-LEAF ELECTROSCOPE

Structure:

- Metal rod with a metal disc at the top.
- Gold leaves attached inside a glass jar.
- Earth wire connected to discharge excess charge.

Working Principle:

- When a charged body touches the disc, charge travels down and spreads to the gold leaves, making them diverge due to repulsion.

Uses:

- Detect presence and type of electric charge.
- Determine quantity of charge (roughly).

- Test insulation of materials.

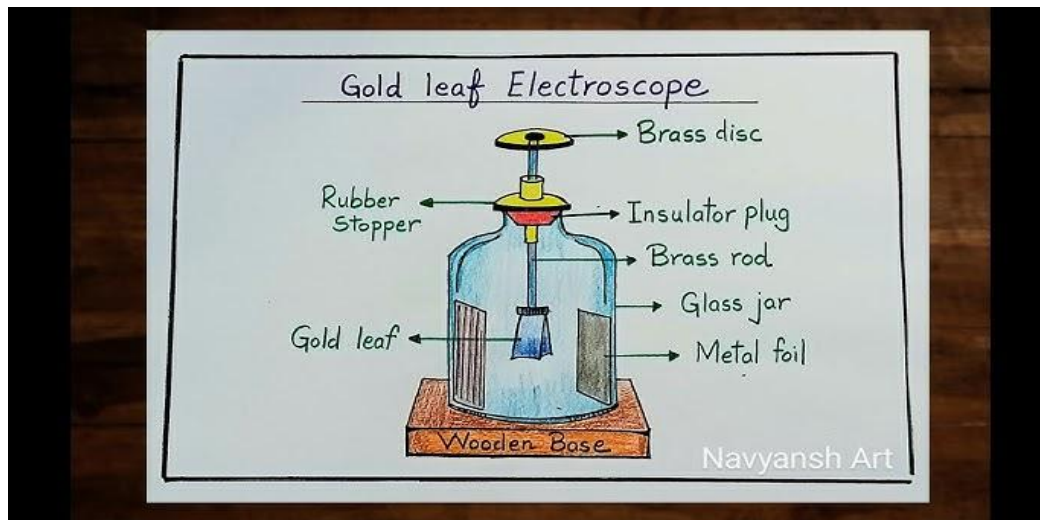


Diagram of a labeled gold-leaf electroscope with diverging leaves.

REAL-LIFE APPLICATIONS OF ELECTRIC CHARGES

1. **Photocopiers and laser printers** – work by charging specific areas of paper.
2. **Spray painting** – ensures even coating by using opposite charges.
3. **Air purifiers and chimney stacks** – remove particles using electric charges.
4. **Lightning** – results from discharge of built-up static charge in clouds.
5. **Electrostatic precipitators** – reduce pollution by removing dust from gases.

EVALUATION

1. The minimum charge a body can have is ____
D. $6.022 \times 10^{23} \text{ C}$ C. 9.8 N/kg B. $3.0 \times 10^8 \text{ C}$ A. $1.6 \times 10^{-19} \text{ C}$
2. Define electric charge and give its SI unit.
3. List and explain three methods of charging a body.
4. State and explain the law of conservation of charge.
5. With the aid of a diagram, explain how an electroscope works.
6. Calculate the total charge carried by 4.0×10^{19} electrons.

Solution to Q5:

Given:

- Number of electrons = 4.0×10^{19}
 - Charge of one electron = $1.6 \times 10^{-19} \text{ C}$
- $$q = n \times e = 4.0 \times 10^{19} \times 1.6 \times 10^{-19} = 6.4 \text{ C}$$

ASSIGNMENT

1. With a well-labeled diagram, describe the working of a gold-leaf electroscope.
2. List five differences between conductors and insulators.
3. Give two examples each of materials that become positively and negatively charged when rubbed.



WEEK NINE - DESCRIPTION AND PROPERTIES OF FIELDS

MEANING OF FIELD IN PHYSICS

In physics, a **field** is a region of space where an object experiences a force due to the presence of another object without physical contact.

Key Points:

- Fields explain **action at a distance**.
- Fields may be **scalar** or **vector** quantities.
 - **Scalar field:** Only magnitude (e.g., temperature field).
 - **Vector field:** Has both magnitude and direction (e.g., gravitational, electric, and magnetic fields).

TYPES OF FIELDS

1. Gravitational Field

- Source: Mass.
- Acts on: Other masses.
- Nature of Force: Always attractive.
- Example: Earth's gravitational pull on objects.

2. Electric Field

- Source: Electric charge.
- Acts on: Other electric charges.
- Nature of Force: Attractive or repulsive depending on the charges.
- Example: Electrostatic force between charged balloons.

3. Magnetic Field

- Source: Moving charges or magnets.
- Acts on: Magnetic materials or moving charges.
- Nature of Force: Can attract or repel.
- Example: Interaction between two bar magnets or between a current-carrying wire and a compass needle.
-

REPRESENTATION OF FIELDS

1. Field Lines (Lines of Force)

- Imaginary lines used to represent the direction and strength of the field.
- **Direction:** Shows the direction of force on a test object.
- **Spacing:** The closer the lines, the stronger the field.
- **Important Rules:**
 - Field lines never cross.
 - Field lines originate and terminate based on source:
 - **Electric Field:** From positive to negative.
 - **Gravitational Field:** Toward mass.
 - **Magnetic Field:**² From North to South outside the magnet.

VECTOR REPRESENTATION

- Each point in the field has a vector showing the direction and magnitude of the force.
- Useful in computational modeling.

Field Strength

Field strength is defined as the **force experienced by a unit quantity** placed in the field.

6. Gravitational Field Strength (g)

$$g = \frac{F}{m} = \frac{GM}{r^2}$$

$$G = 6.674 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

- M: Mass producing the field.
- r: Distance from the source mass.
- Unit: N/kg

7. Electric Field Strength (E)

$$E = \frac{F}{q} = \frac{kQ}{r^2}$$

- $k = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
- Q: Source charge.

- Unit: N/C or V/m

8. Magnetic Field Strength (B)

$$F = BIL\sin\theta \Rightarrow B = \frac{F}{IL\sin\theta} \text{ or } B = \frac{F}{qv\sin\theta}$$

- Unit: Tesla (T)
- I: Current, L: Length of conductor, θ : Angle to magnetic field.

PROPERTIES OF FIELDS

Property	Gravitational Field	Electric Field	Magnetic Field
Source	Mass	Charge	Magnet or current
Nature	Always attractive	Attractive/Repulsive	Attractive/Repulsive
Field lines	Point inward	+ve to -ve	N to S (outside)
Force type	Conservative	Conservative	Non-conservative
Affected by	Mass	Charge	Magnetic material
Medium effect	Weak	Strong	Depends on medium
Obeys inverse square law	Yes	Yes	Yes (in some cases)
Can cancel	Yes	Yes	Yes

Inverse Square Law

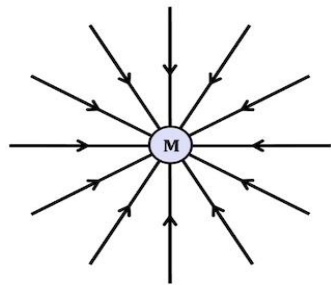
- This is the general law that applies to gravitational and electric fields: This implies that doubling the distance reduces the field strength by a factor of 4.

$$\text{Field strength } E = \frac{F}{q} = \frac{kQ}{r^2}$$

SUPERPOSITION OF FIELDS

- When multiple fields act at a point, the resultant field is the **vector sum** of individual fields.
- Important in:
 - Electric dipoles.
 - Magnetic field interaction.
 - Multiple gravitational sources.

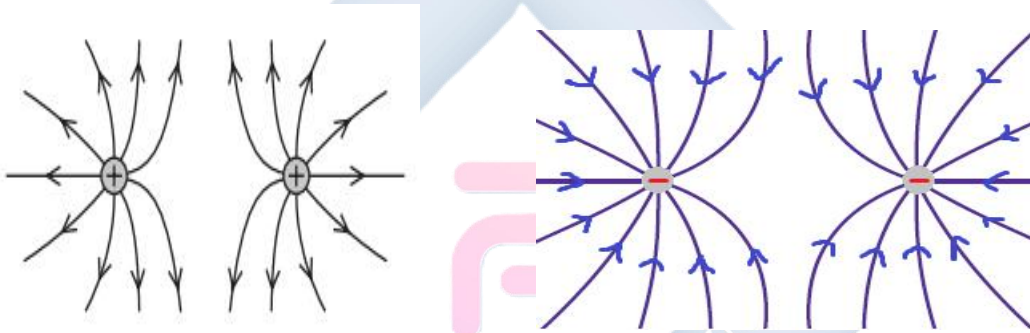
1. Gravitational Field around a Mass



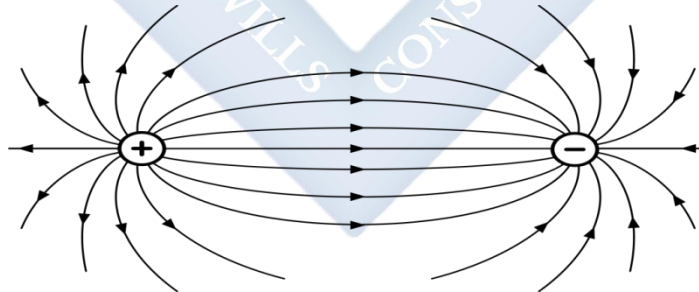
The gravitational field around M is radial

2. Electric Field Between Two Charges

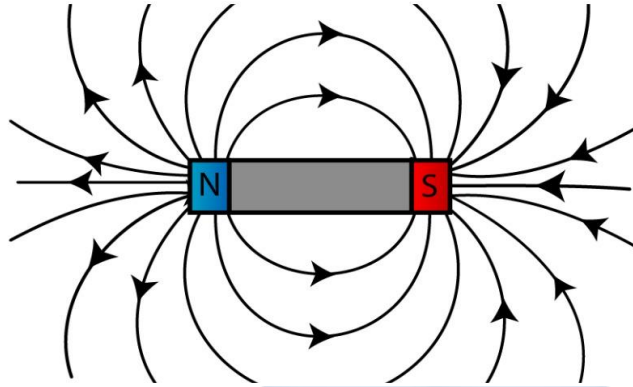
- Like charges:



- Unlike charges:



3. Magnetic Field Around a Bar Magnet



SOLVED EXAMPLES

Example 1:

What is the gravitational field strength of 4m from a mass of 10kg?

Solution

To calculate the **gravitational field strength** at a distance of $r = 4\text{m}$ from a **mass** $M = 10\text{kg}$, we use the formula:

$$g = \frac{F}{m} = \frac{GM}{r^2}$$

Given that: g = Gravitational field strength (N/kg), $G = 6.674 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$, $M = 10\text{kg}$, $r = 4\text{m}$

$$g = \frac{6.674 \times 10^{-11} \times 10}{4^2} = \frac{6.674 \times 10^{-10}}{16} = 4.17125 \times 10^{-11} \text{ N/kg}$$

So, the **gravitational field strength** at 4 meters from a 10 kg mass is approximately $4.17 \times 10^{-11} \text{ N/kg}$.

Example 2:

Find the electric field strength at 2m from a charge of 5.0×10^{-6} C.

Solution

$$E = \frac{F}{q} = \frac{kQ}{r^2}$$

$$= \frac{8.99 \times 10^9 \times 5.0 \times 10^{-6}}{2^2} = \frac{8.99 \times 10^9 \times 5.0 \times 10^{-6}}{4} = \frac{44.9 \times 10^3}{4} = 11,237.5 \text{ N/C}$$

Given that: E = Electric field strength (N/C), $k = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$ (Coulomb's constant), $Q = 5.0 \times 10^{-6}$ C,

Example 3

The force experienced by an object of mass 60.0 kg in the moon's gravitational field is 1.002 x 102 N. What is the intensity of the gravitational field?

Solution

$$\text{Intensity of gravitational field} = \frac{\text{force}}{\text{mass}}$$

$$= \frac{1.002 \times 102 \text{ N}}{60.0 \text{ kg}}$$

$$= 1.67 \text{ Nkg}^{-1}$$

EVALUATION**Objective Questions**

1. Which of these fields can be both attractive and repulsive?

- A. Gravitational
- B. Electric
- C. Magnetic
- D. Both B and C

Answer: D

2. The gravitational field strength on Earth's surface is approximately:

- A. 10 N/kg
- B. 9.8 N/kg
- C. 6.7 N/kg
- D. 8.0 N/kg

Answer: B

Theory Questions

1. Define a field. List and explain three types of fields.
2. Compare gravitational, electric, and magnetic fields using four criteria.
3. With diagrams, describe the field lines around:
 - a. A point charge
 - b. A dipole
 - c. A bar magnet
4. Solve: A force of 2 N acts on a 0.5 C charge. Find the electric field strength.

ASSIGNMENT

1. Draw and label field lines:
 - Between two like electric charges.
 - Between two opposite charges.
 - Around a bar magnet.
2. State five characteristics of each type of field discussed.

WEEK TEN - GRAVITATIONAL FIELD

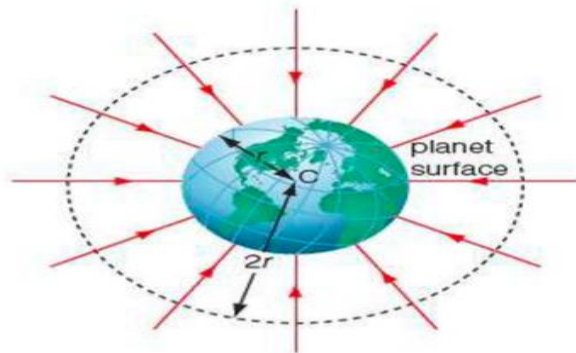
DEFINITION OF A GRAVITATIONAL FIELD

A **gravitational field** is defined as a region of space surrounding a mass in which another mass experiences a force of gravitational attraction.

- It is a **vector field**: It has both **magnitude** and **direction**.
- The direction of the gravitational field at a point is always **toward the mass** producing the field.
- It exists around all masses, but is only noticeable around massive bodies like Earth, planets, and stars.

CONCEPT OF FIELD LINES

- Gravitational fields are represented using **field lines** or **lines of force**.
- These lines:
 - Always point **inward** toward the center of mass.
 - Are **radial** for spherical bodies.
 - Never cross one another.
 - Are denser near the surface (indicating stronger fields).



Gravitational Field Lines Around a Spherical Body

Arrows = Direction of the field (toward the mass center)

NEWTON'S LAW OF UNIVERSAL GRAVITATION

“Every particle in the universe attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centers.”

Mathematical Form:

$$F = \frac{GM_1M_2}{r^2}$$

Where:

- F: gravitational force between two bodies (N)
- m_1, m_2 : masses of the two objects (kg)
- r: distance between the centers of the two bodies (m)
- G: universal gravitational constant = $6.674 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$

GRAVITATIONAL FIELD STRENGTH (G)

The **gravitational field strength** at a point is the **force per unit mass** experienced by a small test mass placed at that point.

$$g = \frac{F}{m} = \frac{GM}{r^2}$$

$$G = 6.674 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

- M: Mass producing the field.
- r : Distance from the source mass.
- Unit: N/kg

At the Surface of the Earth:

$$g = \frac{GM_E}{r^2 R_E} \cong 9.8 \text{ m/s}^2$$

Where:

- $M_E = 5.98 \times 10^{24} \text{ kg}$
- $R_E = 6.37 \times 10^6 \text{ m}$

RELATIONSHIP BETWEEN MASS, WEIGHT, AND G

- **Mass (m):** Scalar quantity; amount of matter; unit: kg.
- **Weight (W):** Vector quantity; gravitational force on a mass:

$$W = mg$$

Variation Of G With Altitude And Depth

1. With Altitude (Above Earth's Surface):

$$g_h = \frac{Gm}{(R+h)^2}$$

Where:

- h: altitude above Earth's surface.
- R: Earth's radius.

As h increases, g decreases.

2. With Depth (Below Earth's Surface):

$$g_d = g_o \left(1 - \frac{d}{R}\right)$$

Where:

- g_d : gravitational field strength at depth d
- g_o : field strength at the surface
- R: Earth's radius

As d increases, g decreases linearly.

Step 6: GRAVITATIONAL POTENTIAL ENERGY (U)

This is the energy a mass possesses due to its position in a gravitational field:

$$U = -\frac{Gm}{r}$$

Negative sign indicates the field is **attractive**.

- Work must be done to separate masses (i.e., increase r).

Properties of Gravitational Fields

Property	Explanation
Vector Field	Has both magnitude and direction (towards mass).
Conservative	Work done is path-independent; energy conserved.
Always Attractive	Gravitational force only pulls, never pushes.
Obeys Inverse Square Law	Field strength decreases with square of distance.
Infinite Range	Extends to infinity, though it weakens with distance.
Superposition Applies	Net field at a point is vector sum of individual fields.

Step 7: Applications of Gravitational Field Concept

- Weight determination.
- Satellite motion and orbiting.
- Planetary motion and space travel.
- Free fall and projectile motion analysis.
- Tides caused by Moon's gravitational pull.
- Structure of stars and galaxies.

Solved Examples

Example 1:

The earth's gravitational field intensity at its surface is about

($G = 6.7 \times 10^{-11} \text{Nm}^2/\text{kg}^2$, mass of the earth is $6 \times 10^{24} \text{kg}$, radius of the earth is $6.4 \times 10^6 \text{m}$, g on the earth = 9.8m/s^2)

Solution

$$E = \frac{F}{m} = \frac{GMm/r^2}{m} = \frac{GM}{r^2}$$

$$E = \frac{6.7 \times 10^{-11} \times 6 \times 10^{24}}{(6.4 \times 10^6)^2} = 9.8 \text{N/kg}$$

$$E = g = 9.8 \text{N/kg}$$

Example 2:

Calculate the force of attraction between a star of mass 2.00×10^{30} kg and the earth assuming the star is located 1.50×10^8 km from the earth.

[Mass of the earth = 5.98×10^{24} kg, $G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$, $g = 10 \text{ ms}^{-2}$]

Solution

$$\text{Using } F = \frac{Gm_em}{r^2}$$

Substituting,

$$F = \frac{6.67 \times 10^{-11} \times 5.98 \times 10^{24} \times 2 \times 10^{30}}{(1.5 \times 10^8)^2}$$

$$= 3.5455 \times 10^{28} \text{ N}$$

Example 3:

The force of attraction between two point masses is 10^{-4} N when the distance between them is 0.18 m. If the distance is reduced to 0.06 m, calculate the force.

Solution

$$F \propto \frac{1}{r^2}$$

$$\frac{F_1}{F_2} = \frac{r_2^2}{r_1^2}$$

$$\frac{10^{-4}}{F_2} = \frac{(0.06)^2}{(0.18)^2}$$

$$F_2 = \frac{10^{-4} \times 0.182}{0.062}$$

$$F_2 = 9 \times 10^{-4} \text{ N}$$

Evaluation Questions

1. Gravitational force between two bodies is:

- A. Repulsive
- B. Attractive
- C. Either
- D. Zero

Answer: B

2. Which quantity remains constant irrespective of location?

- A. Weight
- B. Mass
- C. Gravitational force
- D. Gravitational potential energy

Answer: B

3. If the distance between two bodies is halved, the gravitational force becomes:

- A. Doubled
- B. Four times
- C. Half
- D. Quarter

Answer: B

Theory Questions

1. Define gravitational field strength and derive its expression.
2. State Newton's Law of Universal Gravitation and explain the significance of each term.
3. Describe the variation of gravitational field with altitude.
4. A 50 kg body is placed at a point where the gravitational field strength is 9.8 N/kg. Calculate its weight.

Assignment

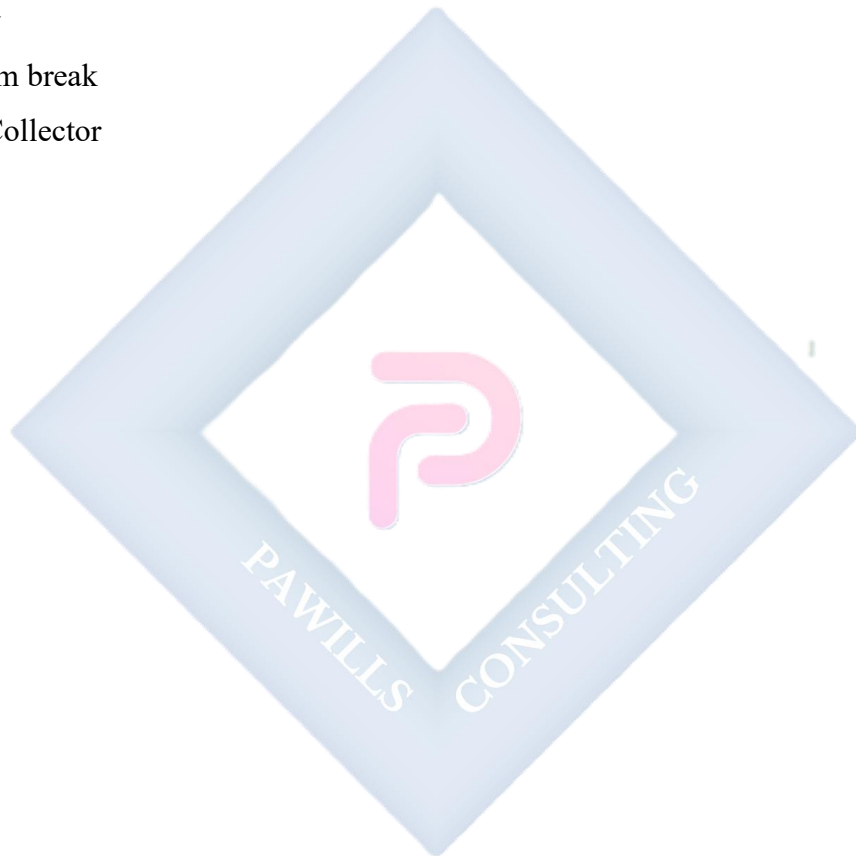
1. A satellite is orbiting at 400 km above Earth's surface. Calculate the gravitational field strength at that height.
(Use $R_E = 6.37 \times 10^6$ m, $G = 6.674 \times 10^{-11}$ Nm²/kg²)
2. A man weighs 800 N on Earth. What will be his weight on the Moon if the gravitational field on the Moon is 1.63 N/kg?



SS2 PHYSICS LESSON NOTES

SECOND TERM

1. Electric Field
2. Magnetic Fields
3. Electro-magnetic Field
4. Simple A.C. Circuits
5. Midterm examination
6. Battery
7. Midterm break
8. Solar Collector



WEEK 1 - ELECTRIC FIELD

MEANING OF ELECTRIC FIELD

An **electric field** is a region of space around a charged body where **another charge** experiences a force.

It is a **vector field**, because it has both **magnitude** and **direction**.

Key Terms:

- **Source Charge (Q):** The charge creating the electric field.
- **Test Charge (q):** A small charge placed in the field to detect its presence.

DIRECTION OF ELECTRIC FIELD

- For a **positive charge**, the electric field radiates **outward**.
- For a **negative charge**, the field lines point **inward** toward the charge.
- The field direction is defined as the direction of force on a **positive test charge**.

ELECTRIC FIELD STRENGTH (E)

Electric field strength is the **force per unit positive charge** experienced by a small test charge in the field.

$$E = \frac{F}{q}$$

Where:

- E = Electric field strength (N/C or V/m)
- F = Force on test charge (N)
- q = Magnitude of test charge (C)

For a Point Charge:

$$E = \frac{kQ}{r^2}$$

Where:

- Q: Source charge (C)
- r: Distance from the charge (m)
- $k = 14\pi\epsilon_0 = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
- ϵ_0 : Permittivity of free space = $8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$

Coulomb's Law

The law states that the force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.

$$F = \frac{kq_1q_2}{r^2}$$

Where:

- q = Two point charges (C)
- r = Separation between charges (m)
- F = Force between them (N)

Nature of Force:

- **Attractive** if charges are opposite.
- **Repulsive** if charges are like.

Properties of Electric Fields

Property	Explanation
Vector Field	Has both magnitude and direction.
Obeys Inverse Square Law	Strength decreases as distance increases.
Direction	Defined as direction of force on a positive test charge.
Conservative Field	Work done is path-independent.
Fields Combine by Superposition	Total field is vector sum of all individual fields.
Strength Depends on Medium	Weaker in insulating materials than in vacuum.

REPRESENTATION OF ELECTRIC FIELDS

Electric Field Lines (Lines of Force):

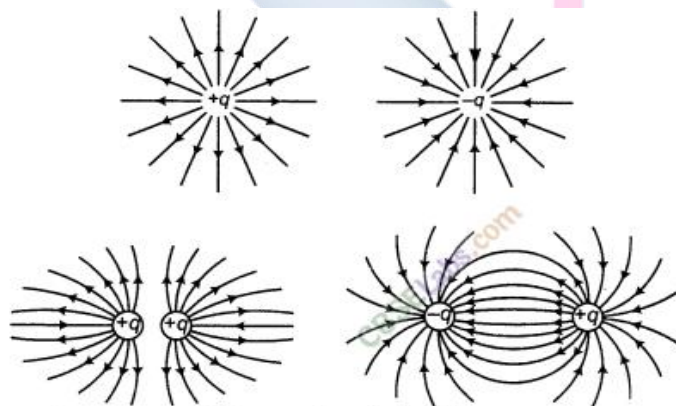
- Imaginary lines that indicate the direction of the electric field.
- Properties:
 1. Begin on **positive** charges and end on **negative** charges.
 2. Never intersect.
 3. The **closer** the lines, the **stronger** the field.
 4. Field lines are **perpendicular** to the surface of a conductor.

Patterns of Electric Field

1. Positive Point Charge

→ Lines radiate **outward**

2. Negative Point Charge



Different electric field lines

→ Lines point **inward**

3. Two Like Charges

→ Lines repel and curve away from each other

4. Two Opposite Charges (Dipole)

→ Lines originate on the positive charge and terminate on the negative charge, forming a curved pattern.

UNIFORM AND NON-UNIFORM ELECTRIC FIELDS

- **Uniform Electric Field:**
 - Field strength is constant in magnitude and direction.
 - Created between **two parallel plates**.
 - Field lines are **parallel and equally spaced**.
- **Non-uniform Electric Field:**
 - Field strength varies with position.
 - Found around **point charges** or irregular charge distributions.

Electric Field in a Medium

The presence of a medium **reduces** the electric field strength compared to vacuum.

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

Where $\epsilon = \epsilon_r$: Relative permittivity (dielectric constant)

CONDUCTORS AND INSULATION

Conductor: These are those materials through which electricity can flow easily. Examples are all metals (such as silver, iron, aluminum, zinc, brass, copper etc.), water, inorganic acid solutions, damp air, etc.

Insulator: These are those materials which do not allow electric current to flow through them easily. Examples are ebonite, glass, dry wood, plastics, organic acids solution etc.

Resistance: This is the opposition to the flow of electric current. Its S. I. Units is in ohm (Ω). A component that is designed to over a certain amount of resistance in a circuit is called a *resistor*.

Arrangements/Connection of Resistor in an Electric Circuit

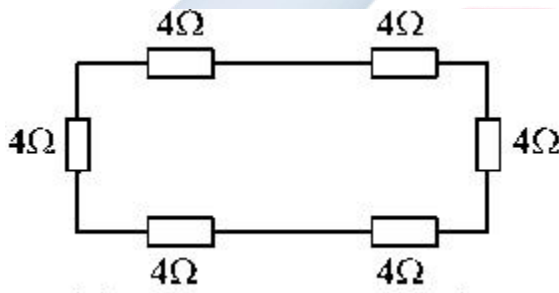
1. Series connection: This is when two resistors are connected in a circuit such that there are connected end to end and the same current passes through each them.

Mathematically, $R = R_1 + R_2 + R_3 + \dots R_n$

2. Parallel connection: This is when two or more resistors are connected side by side in an electric circuit such that their corresponding ends are joined together at two common junctions. The amount of current passes through each resistors is inversely proportional to the value of its resistance.

Mathematically, $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots \frac{1}{R_n}$

Example 1:



Find the effective resistance in the diagram above.

Solution

From the diagram given, the same current will flow across the resistors, hence, they are in series.

$$R_e = 4 + 4 + 4 + 4 + 4 + 4 = 24\Omega$$

Example 2:

A $35\text{ k}\Omega$ is connected in series with a resistance of $40\text{ k}\Omega$. What resistance R must be connected in parallel with the combination so that the equivalent resistance is equal to $25\text{ k}\Omega$?

Solution

For The combinations in series

$$\Rightarrow R_1 = 35\text{ k}\Omega + 40\text{ k}\Omega = 75\text{ k}\Omega$$

R Is Combined with $75\text{ k}\Omega$ in parallel to give $25\text{ k}\Omega$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{R_1}$$

$$\Rightarrow \frac{1}{25} = \frac{1}{R} + \frac{1}{75}$$

$$\Rightarrow \frac{1}{25} - \frac{1}{75} = \frac{1}{R}$$

$$\frac{3 - 1}{75} = \frac{1}{R} \Rightarrow \frac{2}{75} = \frac{1}{R}$$

$$\Rightarrow \frac{75}{2} = R$$

$$\therefore R = 37.5 \text{ k}\Omega$$

SIMILARITIES AND DIFFERENCES WITH GRAVITATIONAL FIELDS

Feature	Electric Field	Gravitational Field
Source	Electric charge	Mass
Nature	Attractive or repulsive	Always attractive
Inverse Square Law	Yes	Yes
Field Lines	Positive to Negative	Toward mass
Test Object	Positive charge	Mass
Constant	$k=4\pi\epsilon_0 \quad 1$	G

Applications of Electric Fields

1. **Cathode ray tubes** – Electric field deflects electron beams.
2. **Photocopiers and laser printers** – Use static electric fields.
3. **Electrostatic precipitators** – Remove particles from exhaust.
4. **Capacitors** – Store energy in electric fields.
5. **Particle accelerators** – Use electric fields to accelerate particles.

Worked Examples

Example 1:

Find the electric field strength 0.1 m away from a point charge of $2.0 \times 10^{-6} \text{ C}$.

Solution

$$E = \frac{Q}{4\pi\epsilon_0 r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-6}}{(0.1)^2} = 1.8 \times 10^5 \text{ N/C}$$

Example 2:

What is the electric field, E, experienced by a charge of magnitude 4 nC at a point where the electric force is 5×10^{-4} N in the x direction?

Solution

$$q = 4 \text{ nC} = 4 \times 10^{-9} \text{ C}; F = 5 \times 10^{-4} \text{ N}; E = ?$$

$$F = qE$$

$$\Rightarrow E = \frac{F}{q} = \frac{5 \times 10^{-4}}{4 \times 10^{-9}}$$

$$\therefore E = 1.25 \times 10^5 \text{ NC}^{-1}$$

Example 3:

Calculate the electric field intensity in vacuum at a distance of 20cm from a charge of 6×10^{-5} C.

$$\left(\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{C}^{-2} \right)$$

Solution:

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$E = \frac{9 \times 10^9 \times 6 \times 10^{-5}}{0.2^2}$$

$$E = 1.35 \times 10^7 \text{ NC}^{-1}$$

EVALUATION

1. Which of the following best defines electric field strength?

- A. Charge per unit mass
- B. Force per unit charge
- C. Force per unit distance
- D. Work per unit mass

Answer: B

2. The SI unit of electric field strength is:

- A. N/m
- B. J/C
- C. N/C
- D. C/N

Answer: C

3. Electric field lines:

- A. Cross each other
- B. End on positive charges
- C. Start on negative charges
- D. Never cross each other

Answer: D

Theory Questions

1. Define electric field and state its unit.
2. Derive the formula for the electric field due to a point charge.
3. A $2\ \mu\text{C}$ charge experiences a force of $0.01\ \text{N}$ in a field. Find the field strength.
4. Compare gravitational and electric fields (at least four points).

ASSIGNMENT

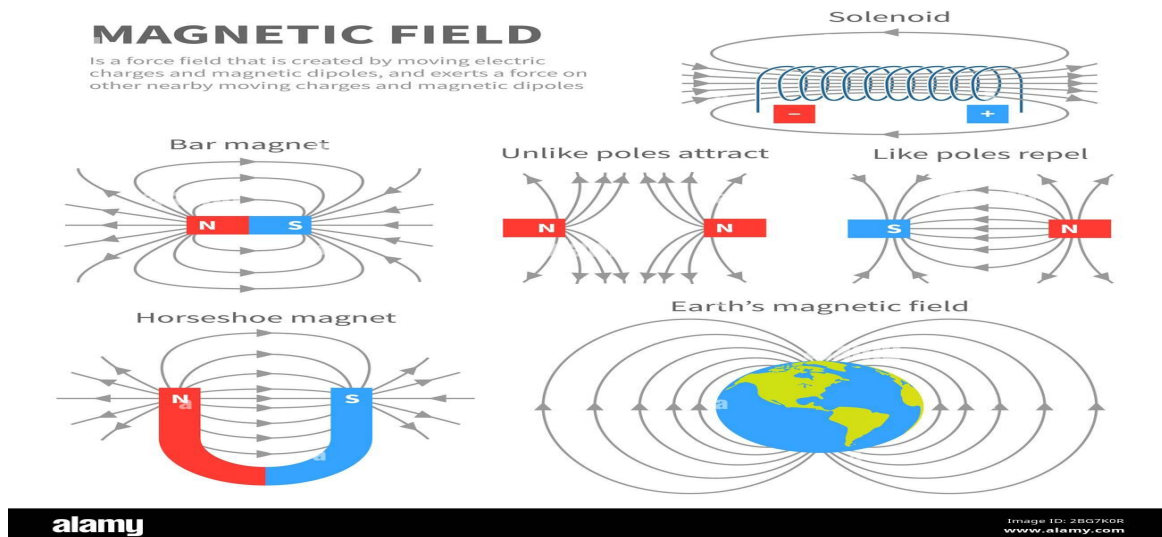
1. Draw field line diagrams for:
 - a. A positive point charge

- b. A dipole system
 - c. Two like charges
2. A test charge of $4\mu\text{C}$ experiences a force of $2 \times 10^{-2} \text{ N}$. Calculate the electric field at that point.
3. Two point charges of $6\mu\text{C}$ and $8\mu\text{C}$ are placed 3 m apart. Find the force between them.

WEEK 2 - MAGNETIC FIELDS

MEANING OF MAGNETIC FIELD

A **magnetic field** is defined as the region of space surrounding a magnet, current-carrying conductor, or moving charged particle where a magnetic force is exerted on other magnetic materials or moving charges.



SOURCES OF MAGNETIC FIELD

(i) Permanent Magnets

- Naturally occurring or artificially created magnets with permanent magnetic properties.
- Examples: Bar magnets, horseshoe magnets, neodymium magnets.
- Fields are strongest at poles.

(ii) Current-Carrying Conductors

- When electric current flows through a wire, it generates a magnetic field around it.
- The direction of the field is given by the **Right-Hand Grip Rule**.

(iii) Solenoids and Electromagnets

- A **solenoid** is a long coil of wire; when current flows through it, it acts like a bar magnet.
- **Electromagnets** consist of solenoids with iron cores to enhance the magnetic field.

(iv) Moving Charges

Any moving electric charge produces a magnetic field (basis for electromagnetism).

Uses of electromagnets

- Generators /electric motors
- Separation of iron scraps from debris
- Cranes for transporting heavy magnetic materials
- Electric bells /telephone earpieces
- Industrial robots[in automobile assembly plants]
- Electromagnetic switches [magnetic relays]
- Medical diagnosis [magnetic resonance scanners]

MAGNETIC FIELD LINES (LINES OF FORCE)

This is an imaginary lines used to represent the direction and strength of a magnetic field.

Properties:

1. They originate from the **North pole** and terminate at the **South pole** outside the magnet.
2. They form **closed loops**, continuing through the magnet from south to north internally.
3. Field lines **never intersect** each other.
4. The **density of lines** indicates field strength – closer lines mean stronger fields.
5. Tangent to the field line at any point gives the direction of the field at that point.

6. Magnetic field lines exhibit **tension** (they try to contract) and **repulsion** (they repel each other sideways).

PATTERNS OF MAGNETIC FIELD AROUND SOURCES

(a) Bar Magnet:

- Field lines are strongest at poles.
- Shape: Symmetrical loops from North to South outside the magnet.

(b) Straight Current-Carrying Wire:

- Field lines form concentric circles around the wire.
- Use **Right-Hand Grip Rule**: Thumb points in current direction, fingers curl in field direction.

(c) Circular Loop of Wire:

- Field lines are circular, but stronger and denser at the center of the loop.

(d) Solenoid:

- Resembles field of a bar magnet.
- Strong and uniform field inside; weak and divergent outside.
- Polarity depends on direction of current.

RIGHT-HAND RULES

(i) Right-Hand Grip Rule (Straight Wire):

- Thumb: Direction of current
- Fingers: Direction of magnetic field (circular)

(ii) Right-Hand Solenoid Rule:

- Fingers curl in current direction
- Thumb points toward magnetic North pole

MAGNETIC FLUX AND FLUX DENSITY

(i) Magnetic Flux (Φ): This is the total number of magnetic field lines passing through a surface.

$$\Phi = B.A.\cos\theta$$

Where:

- Φ = magnetic flux (Wb)
- B = magnetic flux density (T)
- A = area (m^2)
- θ = angle between B and A

(ii) Magnetic Flux Density (B):

- Magnetic flux per unit area.

$$B = \frac{\Phi}{A.\cos\theta}$$

SI Unit: **Tesla (T)** = Wb/m^2

FORCE ON CURRENT-CARRYING CONDUCTOR IN A MAGNETIC FIELD

When a conductor carrying current is placed in a magnetic field, it experiences a force. This is the **motor effect**, and it is the working principle behind electric motors.

Formula (Lorentz Force):

$$F = BIL\sin\theta$$

Where:

- F = force (N)
- B = magnetic field strength (T)
- I = current (A)
- L = length of conductor (m)
- θ = angle between conductor and field

Direction of Force:

- Determined using **Fleming's Left-Hand Rule:**
 - First Finger: Field (B)
 - Second Finger: Current (I)
 - Thumb: Force (Motion)

MAGNETIC FIELD OF A CIRCULAR LOOP

The magnetic field at the center of a single circular coil carrying current is given by:

$$B = \mu_0 I^2 R$$

For a coil with N turns: $B = \mu_0 N I^2 R$

Where:

- μ = Permeability of free space ($4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$)
- R = radius of the coil

INTERRELATIONSHIP BETWEEN ELECTRICITY AND MAGNETISM

- A changing electric current produces a magnetic field.
- A changing magnetic field can induce an electric current (**Electromagnetic Induction**).
- This relationship is central to **electromechanical devices** such as generators, motors, transformers.

WORK EXAMPLES

Example 1:

A wire of length 10cm carrying a current of 4.0A is placed between the poles of a powerful electromagnetic flux density 2.0T. Calculate the force on the wire when it makes an angle 60° with the field.

Solution

From the parameters given:

$$F = BIL \sin \theta = 2 \times 4 \times 0.1 \sin 60^\circ = 0.8 \times 0.8660 = 0.6928 = 0.69 \text{ N}$$

Example 2:

A wire of length 10cm carrying a current of 4.0A is placed between the poles of a powerful electromagnetic flux density 2.0T. Calculate the maximum force on the wire

Solution

From the parameters given,

$$F = BIL = 2 \times 4 \times 0.1 = 8 \times 0.1 = 0.8\text{N}$$

APPLICATIONS OF MAGNETIC FIELDS

1. **Electric Motors:** Convert electrical energy to mechanical energy.
2. **Generators:** Convert mechanical energy to electrical energy.
3. **Transformers:** Step up/down AC voltages via mutual induction.
4. **Magnetic Resonance Imaging (MRI):** Used in medical diagnostics.
5. **Speakers and Microphones:** Sound waves to electric signals and vice versa.
6. **Magnetic Levitation (MagLev) Trains:** Floating trains using electromagnets.
7. **Credit/Debit Cards:** Use magnetic stripes to store information.
8. **Data Storage Devices:** Hard disks and tapes use magnetic coatings.
9. **Particle Accelerators:** Use magnetic fields to guide charged particles.
10. **Compasses:** Navigation using Earth's magnetic field.

EVALUATION

1. Define a magnetic field and magnetic flux density.
2. Draw the magnetic field pattern around a bar magnet and a straight wire.
3. State and explain the right-hand grip rule.
4. A wire of length 0.5 m carries 10 A of current and is placed at 90° to a 0.3 T magnetic field. Calculate the force on the wire.
5. What are the similarities and differences between a solenoid and a bar magnet?
6. List five practical applications of magnetic fields.
7. How can the strength of an electromagnet be increased?

ASSIGNMENT

- (a) State five differences between permanent magnets and electromagnets.
- (b) Research and describe the construction and working principle of a DC electric motor, highlighting how magnetic fields play a role.



WEEK 3 - ELECTRO-MAGNETIC FIELDS

MEANING OF ELECTROMAGNETIC FIELD

An **electromagnetic field** is a physical field produced by electrically charged objects, consisting of **both electric and magnetic components**, which interact with other charged particles.

Definition:

An electromagnetic field is the region of space where both electric and magnetic forces act simultaneously due to the presence of moving electric charges (such as electric currents).

COMPONENTS OF AN ELECTROMAGNETIC FIELD

1. Electric Field (E):

- A field produced by stationary electric charges.
- Direction: From positive to negative charges.
- Unit: Volt per meter (V/m)

2. Magnetic Field (B):

- Produced by moving charges (current).
- Direction: Given by the right-hand rule.
- Unit: Tesla (T)

These two fields are **mutually perpendicular** to each other and to the direction of propagation in electromagnetic waves.

ORIGIN OF ELECTROMAGNETIC FIELD

(a) From Moving Charges:

- A **stationary charge** creates only an **electric field**.
- A **moving charge** (current) creates both **electric and magnetic fields**, i.e., an **electromagnetic field**.

(b) From Accelerated Charges:

- When charges **accelerate**, they emit **electromagnetic radiation** (e.g., radio waves, X-rays).

DYNAMIC NATURE OF ELECTROMAGNETIC FIELD

According to **Maxwell's Equations**, a **changing electric field** produces a **magnetic field**, and vice versa.

- This interaction gives rise to **self-propagating electromagnetic waves**, such as light.

PROPERTIES OF ELECTROMAGNETIC FIELDS

1. Consists of oscillating electric and magnetic fields.
2. Fields are perpendicular to each other.
3. Fields propagate together in a direction **perpendicular** to both electric and magnetic fields.
4. Travel at the speed of light in vacuum ($c = 3 \times 10^8 \text{ m/s}$)
5. Can exert forces on other charges (via **Lorentz force**).
6. Can carry energy and momentum (e.g., in radiation pressure).

ELECTROMAGNETIC WAVE STRUCTURE

- **Electric Field (E)** oscillates in one plane.
- **Magnetic Field (B)** oscillates in a perpendicular plane.
- Direction of wave propagation is **perpendicular** to both E and B.

This structure forms the basis of all forms of electromagnetic radiation: radio, microwave, infrared, visible light, UV, X-rays, and gamma rays.

REPRESENTATION OF ELECTROMAGNETIC FIELD

(i) Right-Hand Rule for Electromagnetic Wave:

- Thumb: Direction of wave propagation

- First finger: Direction of Electric Field (E)
- Second finger (perpendicular): Direction of Magnetic Field (B)

LORENTZ FORCE LAW

The law states that the total force on a moving charge in an electromagnetic field is:

$$F = q(E + v \times B)$$

Where:

- F = electromagnetic force (N)
- q = charge (C)
- E = electric field (V/m)
- v = velocity of the particle (m/s)
- B = magnetic field (T)

ELECTROMAGNETIC INDUCTION

Discovered by **Michael Faraday**, it is the production of **electromotive force (emf)** in a conductor due to a **changing magnetic field**.

Faraday's Law:

$$\text{Emf} = -\frac{d\Phi_B}{dt}$$

Where:

- Φ_B = magnetic flux (Wb)

This is the principle behind:

- Electric generators
- Transformers
- Inductors

DIFFERENCES BETWEEN ELECTRIC, MAGNETIC, AND ELECTROMAGNETIC FIELDS

Feature	Electric Field	Magnetic Field	Electromagnetic Field
Source	Stationary charge	Moving charge (current)	Moving or accelerating charge
Vector quantity	Yes	Yes	Yes
Affects	Charges	Moving charges	Both charges and currents
Produces	Static force	Circular magnetic force	Radiation and propagation of energy
Unit	Volt/meter (V/m)	Tesla (T)	N/A (combo of E and B units)
Exists independently?	Yes	Yes	Not independent — E and B interlinked

EVALUATION QUESTIONS

1. Define an electromagnetic field.
2. Explain how a moving electric charge gives rise to an electromagnetic field.
3. Differentiate between electric field and magnetic field.
4. Describe the direction and structure of an electromagnetic wave.
5. What is the significance of Faraday's Law in electromagnetic fields?
6. List four practical applications of electromagnetic fields.
7. A wire carries a current of 6 A and is perpendicular to a 0.5 T magnetic field. What is the force on 2 m of the wire?

ASSIGNMENT

- (a) State Maxwell's equations in words.
- (b) Describe how electromagnetic waves are generated and propagate.
- (c) Explain the role of electromagnetic fields in mobile phone communication.

WEEK 4 - SIMPLE A.C. CIRCUITS

MEANING OF ALTERNATING CURRENT (A.C.)

Alternating Current (A.C.) is an electric current that **reverses its direction periodically** and varies continuously with time.

- In contrast to **Direct Current (D.C.)**, which flows in one direction only.
- A.C. typically follows a **sinusoidal waveform**:

$$I(t) = I_0 \sin(\omega t)$$

Where:

- I_0 = peak current
- ω = angular frequency ($= 2\pi f$)
- f = frequency (Hz)
- t = time (s)

CHARACTERISTICS OF A.C

- **Frequency (f):** Number of cycles per second (unit: Hz)
- **Amplitude:** Maximum value of current or voltage (peak value)
- **Period (T):** Time taken for one complete cycle ($T = 1/f$)
- **Root Mean Square (RMS) Value:** Equivalent steady D.C. value
 - For current:

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$$

- For voltage:

$$V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$$

STEP 3: COMPONENTS OF SIMPLE A.C. CIRCUITS

(i) Resistor (R):

- **Voltage and current are in phase.**
- Ohm's law applies: $V=IR$
- (ii) Inductor (L):
- Causes current to **lag voltage** by 90° .
- Opposition to current is called **inductive reactance**:

$$X_L = 2\pi fL$$

Where L = inductance (H), f = frequency (Hz)

(iii) Capacitor (C):

- Causes current to **lead voltage** by 90° .
- Opposition to current is called **capacitive reactance**:

$$X_C = \frac{1}{2\pi fC}$$

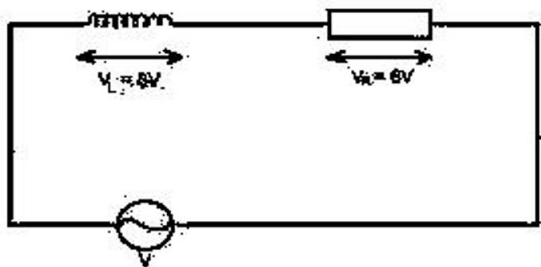
Where C = capacitance (F)

When an a.c voltage of frequency f is applied to a capacitance, c , then

$$V = IX_L$$

COMBINATION OF R, L, AND C IN A.C. CIRCUITS

(a) R-L Circuit:



- Voltage leads current
- Impedance:

$$Z = \sqrt{R^2 + X_L^2}$$

Example 1:

An inductor having an inductance of 120 mH and a resistance of 5 Ω is connected across a 220 V, 60 Hz line. Calculate the power loss in the inductor.

Solution

$L = 120 \text{ mH} = 0.12 \text{ H}$; $R = 5 \text{ } \Omega$; $V = 220 \text{ V}$; $f = 60 \text{ Hz}$; $P_{\text{loss}} = ?$

$$\Rightarrow X_L = 2\pi fL = 2 \times \pi \times 60 \times 0.12$$

$$\Rightarrow X_L = 45.239 \text{ } \Omega$$

$$Z^2 = R^2 + X_L^2$$

$$\Rightarrow Z^2 = 5^2 + 45.239^2$$

$$\Rightarrow Z^2 = 25 + 2046.57$$

$$\Rightarrow Z^2 = 2071.57$$

$$\Rightarrow Z = \sqrt{2071.57}$$

$$\Rightarrow Z = 45.51 \text{ } \Omega$$

$$V = IZ$$

$$\Rightarrow I = \frac{V}{Z} = \frac{220}{45.51}$$

$$Z = 45.51$$

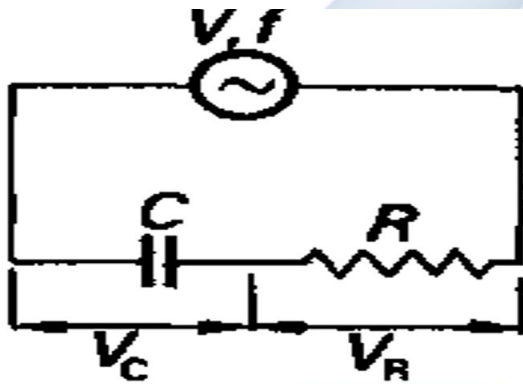
$$\Rightarrow I = 4.83 \text{ A}$$

Power Loss in An inductor is given as:

$$P_{\text{loss}} = I^2 R \text{ (where } R \text{ Is the resistance of the inductor)}$$

$$\therefore P_{\text{loss}} = 4.832^2 \times 5 = 116.82 \text{ W}$$

(a) **R-C Circuit:**



- Current leads voltage
- Impedance:

$$Z = \sqrt{R^2 + X_C^2}$$

Example

An alternating source is connected in series to a capacitor of capacitive reactance $10^3\sqrt{3}$ and a resistor of resistance 10^3 Ohms. The Impedance of the circuit is...

Solution

$$Z = \sqrt{R^2 + X_C^2}$$

$$Z = \sqrt{(10^3\sqrt{3})^2 + (10^3)^2}$$

$$Z = \sqrt{3(10^6) + (10^6)}$$

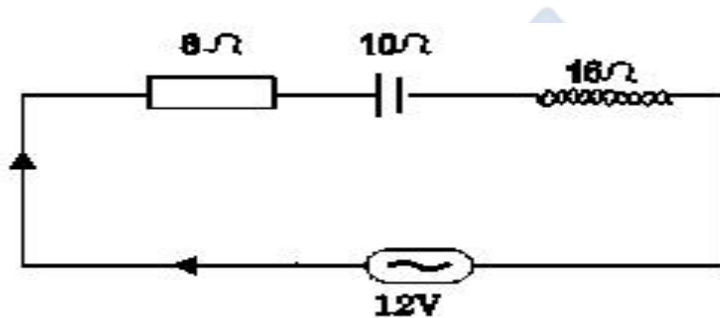
$$Z = \sqrt{(3+1)10^6}$$

$$Z = 2 \times 10^3 \text{ Ohms}$$

$$Z = 2000 \text{ Ohms}$$

(c) R-L-C Circuit (Series):

- Combination of resistor, inductor, and capacitor
- Net reactance:



$$X = X_L - X_C$$

- Impedance:

$$Z = R^2 + (X_L - X_C)^2$$

- Phase angle:

$$\tan \phi = \frac{X}{R}$$

POWER IN A.C. CIRCUITS

Types of Power:

Type	Symbol	Formula	Unit
Apparent Power	S	$S = V_{rms} I_{rms}$	VA
Real Power (Active)	P	$P = VI \cos \phi$	Watt (W)
Reactive Power	Q	$Q = VI \sin \phi$	VAR
Power Factor	—	$\cos \phi = \frac{P}{S}$	Dimensionless

RESONANCE IN A.C. CIRCUITS

Occurs in an RLC circuit when:

$$X_L = X_C \Rightarrow 2\pi fL = \frac{1}{2\pi fC}$$

At resonance:

- Impedance is minimum ($Z = R$)
- Circuit behaves like a pure resistor
- Current is maximum
- Power factor is unity (1)

Resonant Frequency: Resonance frequency in RLC series circuit is the frequencies at which the total impedance of the circuit is minimum or current in the circuit becomes maximum, or the capacitive reactance is equal to the inductive reactance or the impedance is equal to resistance of the resistor.

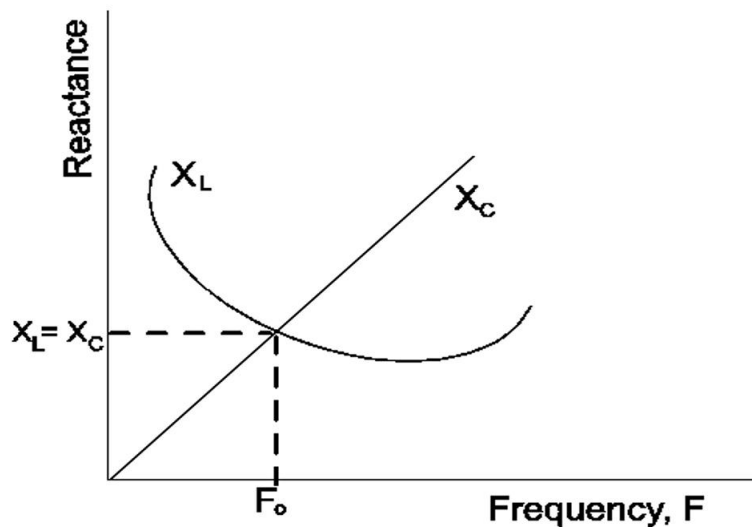


Diagram illustrating the variation of frequency, f , with resistance R . The capacitive reactance, X_C and the inductive reactance, X_L in RLC series circuit.

$$f_0 = \frac{1}{2\pi LC}$$

PHASOR DIAGRAMS

- Phasors are rotating vectors representing sinusoidal voltages/currents.
- Used to show the phase relationships in R, L, C, or RLC circuits.
- In:
 - **Pure R:** V and I in phase
 - **Pure L:** V leads I by 90°
 - **Pure C:** I leads V by 90°
 - **RLC:** Use vector addition to determine resultant voltage/current

APPLICATIONS OF A.C. CIRCUITS

1. Electric Power Distribution (NEPA grid)
2. Transformers(work only with A.C.)
3. Electric Motors (Induction motors)
4. Radio and TV Tuners (resonance in RLC circuits)
5. A.C. Circuit Breakers and Relays
6. Household Appliances (fans, fridges)
7. Signal Processing in Electronics

COMMON A.C. CIRCUIT FORMULAS

$$I = \frac{V}{R}$$

- $Z = \sqrt{R^2 + (X_L - X_C)^2}$
- $X_L = 2\pi fL$
- $X_C = \frac{1}{2\pi fC}$
- $P = VI \cos \phi$
- Power factor = $\cos \phi$

WORK EXAMPLES:

Example 1:

In an electric circuit, an inductor of inductance 0.5 h and resistance 50 Ω is connected to an alternating current source of frequency 60 1b. Calculate the impedance of the circuit.

Solution

Inductance, $l = 0.5\text{H}$, Resistance, $R = 50\Omega$, Frequency, $f = 60\text{Hz}$, impedance, $z = ?$

$$Z = \sqrt{R^2 + X_L^2}$$

Where X_L =inductive reactance

$$X_L = 188.66\Omega$$

$$Z = \sqrt{50^2 + 188.62^2}$$

$$Z = \sqrt{2500 + 3559.96}$$

$$Z = \sqrt{3809.96}$$

$$A = 195.1\Omega$$

$$Z = 195.0\Omega$$

Example 2:

An inductor of inductance 1.0H is connected in series with a capacitor of capacitance 2.0μF in an A.C circuit. Calculate the value of frequency that will make the circuit to resonate.

Solution

Given: Inductance, $L = 1.0\text{H}$

Capacitance, $C = 2.0\mu\text{F}$

Resonance frequency, $f_0 = ?$

1

$$\text{Recall: } f_0 = \frac{1}{2\pi\sqrt{LC}}$$

1

$$\therefore f_0 = \frac{1}{2 \times 3.142 \sqrt{1 \times 2 \times 10^{-6}}}$$

6

1

$$\therefore F_o = \frac{1}{6.282 \times \sqrt{2} \times 10^{-3}}$$

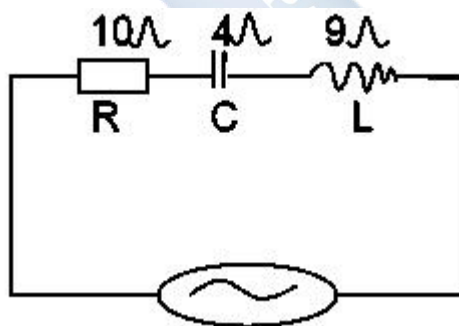
$$F_o = \frac{1}{8.9} \times 10^3 = \frac{1000}{8.9} = 112.5\text{Hz}$$

Example 3:

In a series R-L-C circuit, $R = 10 \text{ ohm}$, $X_c = 4\Omega$ and $X_L = 9 \text{ ohm}$. The impedance of the circuit is

Solution

Diagrammatically, we have



$R \rightarrow$ Resistance {opposition to the flow of current in a resistor}

$X_L \rightarrow$ Inductive reactance {opposition to the flow of current in an inductor}

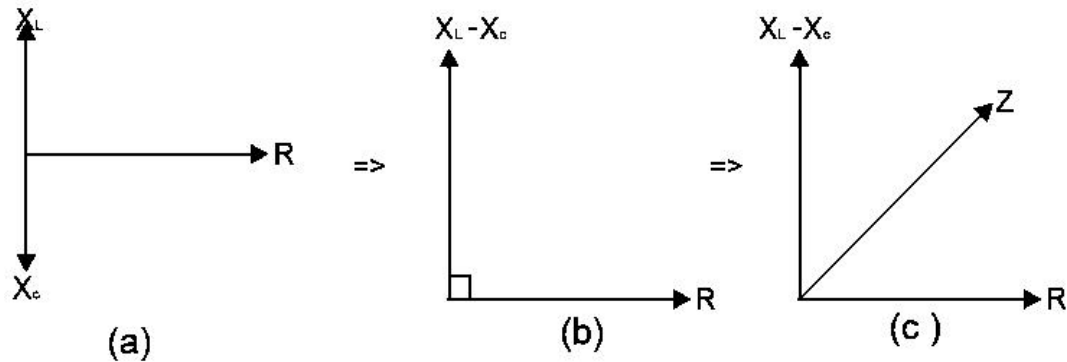
$X_C \rightarrow$ Capacitive reactance {opposition to the flow of current in a capacitor}

$Z \rightarrow$ Impedance {resultant or effective opposition}

to the flow of current}

If you are interested in the phasor diagram, here is it,

We are going to use (c) to compute Z



$$\therefore z^2 = 102 + 52$$

$$z^2 = 100 + 25$$

$$z^2 = 125$$

$$z = \sqrt{125} = 5\sqrt{5} = 11.2 \text{ ohm}$$

EVALUATION

1. Define alternating current.
2. List three differences between A.C. and D.C.
3. Explain what is meant by impedance in an A.C. circuit.
4. In an electric circuit, an inductor of inductance 0.5 h and resistance 50Ω is connected to an alternating current source of frequency 60 lb . Calculate the impedance of the circuit
5. What is resonance in an RLC circuit? State the condition for resonance.
6. A $100 \mu\text{F}$ capacitor is connected to a 50 Hz A.C. supply. Calculate the capacitive reactance.
7. Describe the significance of the power factor in an A.C. circuit.

ASSIGNMENT

(a) With the aid of a diagram, explain what happens in an A.C. circuit containing a resistor, inductor, and capacitor in series.

(b) Discuss five real-life applications of resonance in A.C. circuits.

WEEK 6 – BATTERY

DEFINITION OF A BATTERY

A **battery** is a device that converts **chemical energy into electrical energy** by means of **electrochemical reactions** between two electrodes in an electrolyte.

Note:

- A single unit is called a **cell**.
- A battery is composed of **two or more cells** connected in series or parallel to provide greater voltage or current.

TYPES OF BATTERIES

(i) Primary Batteries (Non-rechargeable):

- Used once and discarded.
- Chemical reaction is irreversible.
- Examples:
 - Dry cell (Leclanché cell)
 - Alkaline cell
 - Mercury cell
 - Zinc-carbon cell

(ii) Secondary Batteries (Rechargeable):

- Can be recharged and reused.
- Chemical reaction is reversible.
- Examples:
 - Lead-acid accumulator
 - Nickel-cadmium cell
 - Lithium-ion battery
 - Nickel-metal hydride (NiMH)

STRUCTURE AND WORKING OF A SIMPLE BATTERY

A simple battery (cell) consists of:

- **Two electrodes** (anode and cathode)
- **Electrolyte**: a chemical solution that facilitates ion flow
- **External circuit**: provides a path for electron flow

Working Principle:

1. Electrochemical reaction occurs between electrodes and electrolyte.
2. Electrons are released at the **anode** and flow through the external circuit to the **cathode**.
3. This movement produces an **electric current**.

ELECTROMOTIVE FORCE (e.m.f.) AND TERMINAL VOLTAGE

(i) Electromotive Force (e.m.f.):

The total energy supplied by the battery per unit charge in driving charges through the entire circuit (external + internal).

$$\text{e.m.f. (E)} = \frac{\text{Work done (W)}}{\text{Charge (Q)}}$$

Unit: Volt (V)

(ii) Terminal Voltage (V):

The actual voltage measured across the battery terminals when it is delivering current.

$$V = E - Ir$$

Where:

- V = terminal voltage
- E = e.m.f.
- I = current

- r = internal resistance of the battery

INTERNAL RESISTANCE OF A BATTERY

- Batteries are not perfect sources of voltage; they have **internal resistance (r)** due to the materials inside the cell.
- It causes a **voltage drop** inside the battery when current flows.
- **Greater internal resistance = more energy lost as heat**

CONNECTION OF BATTERIES

(i) Series Connection:

- Positive terminal of one cell connected to the negative of the next.
- Total e.m.f. = sum of individual e.m.fs

$$E_{\text{total}} = E_1 + E_2 + \dots + E_n$$

(ii) Parallel Connection:

- All positive terminals connected together, and all negative terminals together.
- Voltage remains the same, but current capacity increases.

ADVANTAGES OF BATTERIES

1. Portable and compact power source
2. Quiet and clean operation
3. Useful in remote or mobile environments
4. Rechargeable types reduce waste
5. Available in various sizes and voltages

LIMITATIONS OF BATTERIES

1. Limited capacity and lifespan
2. Environmental pollution (if improperly disposed)
3. Can leak or explode if mishandled
4. Rechargeable batteries lose capacity over time

5. Internal resistance reduces efficiency

APPLICATIONS OF BATTERIES

1. Flashlights and torches
2. Mobile phones and tablets
3. Laptops and other electronics
4. Automobiles (car batteries)
5. Solar energy storage systems
6. Medical devices (e.g. pacemakers)
7. Remote controls and toys
8. Power banks and uninterruptible power supply (UPS)

SAFETY PRECAUTIONS WHEN HANDLING BATTERIES

1. Avoid mixing old and new batteries
2. Do not short-circuit terminals
3. Keep away from heat and fire
4. Store in a cool, dry place
5. Properly recycle or dispose of dead batteries
6. Handle lead-acid batteries with gloves and eye protection
7. Never attempt to recharge non-rechargeable cells

KEY FORMULAS TO REMEMBER

1. $V = E - Ir$
2. $E = \frac{W}{Q}$
3. $R_{\text{total}} (\text{series}) = R_1 + R_2 + \dots + R_n$
4. $R_{\text{total}} (\text{parallel}) = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$

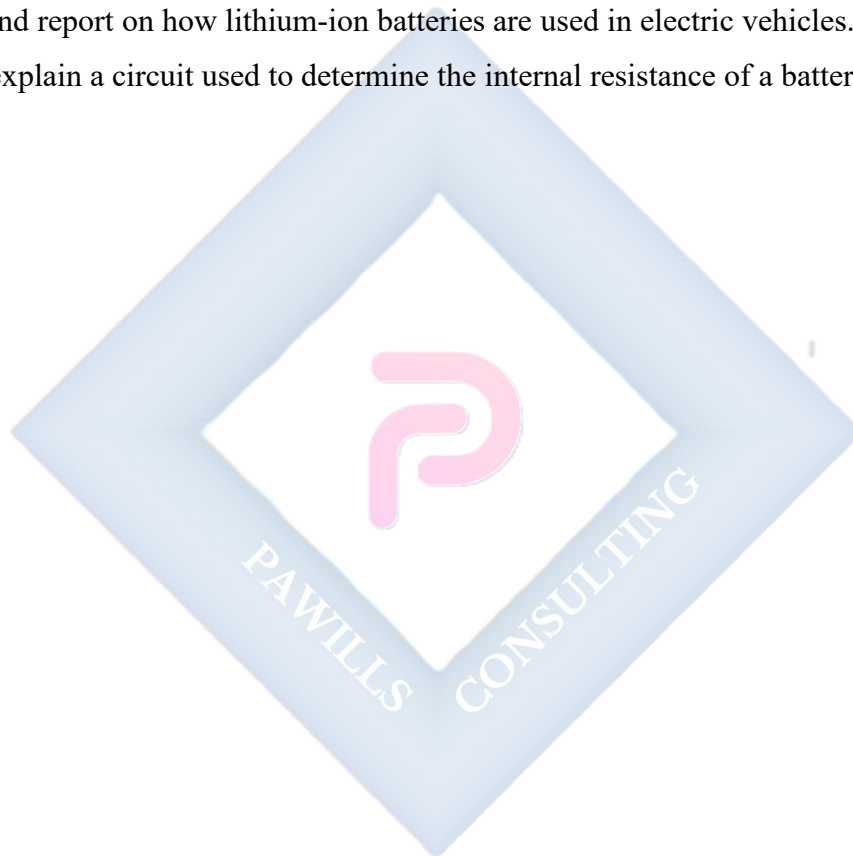
EVALUATION

1. Define a battery.
2. Differentiate between a primary and a secondary cell.
3. What is the e.m.f. of a battery and how is it different from terminal voltage?

4. A battery of e.m.f. 12 V and internal resistance $0.5\ \Omega$ is connected to a $5.5\ \Omega$ resistor.
Calculate the current in the circuit and terminal voltage.
5. Why is internal resistance important in battery design?
6. List four safety precautions when using batteries.

ASSIGNMENT

- (a) With the aid of a diagram, describe the construction and working of a dry cell (Leclanché cell).
- (b) Research and report on how lithium-ion batteries are used in electric vehicles.
- (c) Draw and explain a circuit used to determine the internal resistance of a battery.



WEEK 8 - SOLAR COLLECTOR

DEFINITION OF SOLAR COLLECTOR

A **solar collector** is a device that absorbs solar radiation and converts it into heat energy, which is then used for heating fluids like water or air.

It is a **key component** in **solar thermal energy systems**, as opposed to **photovoltaic panels** that convert sunlight into electricity.

WORKING PRINCIPLE OF A SOLAR COLLECTOR

The basic working principle is based on the **conversion of solar radiation (light energy)** into **thermal energy (heat)**.

Process:

1. **Absorption of sunlight** by a dark surface (absorber plate).
2. **Conversion** of light energy into heat.
3. **Transfer** of this heat to a **working fluid** (air, water, or antifreeze liquid).
4. Heated fluid is then **used directly** (e.g., water heating) or stored in an insulated tank.

MAIN TYPES OF SOLAR COLLECTORS

(i) Flat-Plate Solar Collector

- Most common type for domestic use.
- Consists of an insulated box with a transparent cover and dark-colored absorber plate.
- Heat is transferred to fluid (usually water) passing through tubes attached to the absorber.

(ii) Evacuated Tube Collector

- Series of parallel glass tubes with vacuum between two layers to reduce heat loss.
- More efficient in colder climates.

(iii) Concentrating (Parabolic) Solar Collector

- Uses mirrors or lenses to **focus sunlight** onto a small area.

- High temperatures achieved; used in **industrial or power plant applications**.

COMPONENTS OF A FLAT-PLATE SOLAR COLLECTOR

- 1. Transparent Cover (Glazing):**
 - Allows sunlight in but prevents heat loss.
- 2. Absorber Plate:**
 - Painted black to absorb maximum solar energy.
- 3. Tubes or Channels:**
 - Fluid (water/antifreeze) flows through these to collect heat.
- 4. Insulation Layer:**
 - Prevents heat from escaping from the back and sides.
- 5. Casing (Enclosure):**
 - Protective box for durability and support.
- 6. Storage Tank:**
 - Stores the heated fluid for use when sunlight is not available.
- 7. Pump (optional):**
 - Circulates fluid between collector and storage tank (used in active systems).

COMPARISON: SOLAR COLLECTORS vs SOLAR PANELS

Feature	Solar Collector	Solar Panel (Photovoltaic)
Converts	Sunlight to heat	Sunlight to electricity
Output form	Hot fluid (water or air)	Electric current (DC)
Main application	Water/space heating	Powering electronics/appliances
Efficiency (typical)	~60–80% (thermal)	~15–25% (electrical)
Storage	Thermal tank	Batteries (for electricity)

ADVANTAGES OF SOLAR COLLECTORS

1. Renewable energy source
2. Low operating cost after installation
3. Environmentally friendly – no greenhouse gas emission

4. Quiet operation (no moving parts in passive systems)
5. Ideal for rural and off-grid areas
6. Reduces reliance on fossil fuels
7. Simple maintenance and long lifespan

LIMITATIONS OF SOLAR COLLECTORS

1. Dependent on sunlight – less effective on cloudy/rainy days
2. Initial installation cost can be high
3. Space requirement for large collectors
4. Thermal losses during nighttime if storage is poor
5. Not suitable for portable applications
6. May require anti-freeze fluid in cold climates

APPLICATIONS OF SOLAR COLLECTORS

1. Domestic water heating
2. Swimming pool heating
3. Space heating for homes
4. Industrial process heating
5. Solar desalination (purifying seawater)
6. Greenhouse heating in agriculture
7. Cooking using solar cookers (variant design)
8. Solar drying of crops

TYPES OF SOLAR COLLECTOR SYSTEMS

(a) Passive Systems

- Do not use pumps or controllers.
- Water flows by natural circulation (thermosiphon).
- Simple, low cost, and low maintenance.

(b) Active Systems

- Use **pumps** and **control systems** to circulate fluid.
- More efficient and suitable for larger or multi-storey systems.

DESIGN CONSIDERATIONS FOR EFFICIENCY

1. **Tilt Angle:** Should be equal to the local latitude for optimal absorption.
2. **Orientation:** South-facing (in Northern Hemisphere; North-facing in Southern Hemisphere)
3. **Insulation:** Prevents heat losses from the back and sides.
4. **Material of absorber:** Good thermal conductivity (e.g. copper or aluminum).
5. **Color of absorber:** Black for maximum radiation absorption.

FORMULAE AND CALCULATIONS (Where applicable)

The thermal power Q collected by a solar collector can be approximated by:

$$Q = A \times S \times \eta$$

Where:

- Q = Heat energy collected (W)
- A = Area of collector (m^2)
- S = Solar insolation (W/m^2)
- η = Efficiency (typically 0.6–0.8)

EVALUATION QUESTIONS

1. What is a solar collector?
2. Describe the working principle of a flat-plate solar collector.
3. List and describe three types of solar collectors.
4. Distinguish between solar collector and solar panel.
5. State four advantages and four limitations of solar collectors.
6. What are the main components of a flat-plate solar collector?
7. Calculate the heat collected by a solar panel of area 1.5 m^2 receiving solar radiation at $800 \text{ W}/\text{m}^2$ with 65% efficiency.

ASSIGNMENT

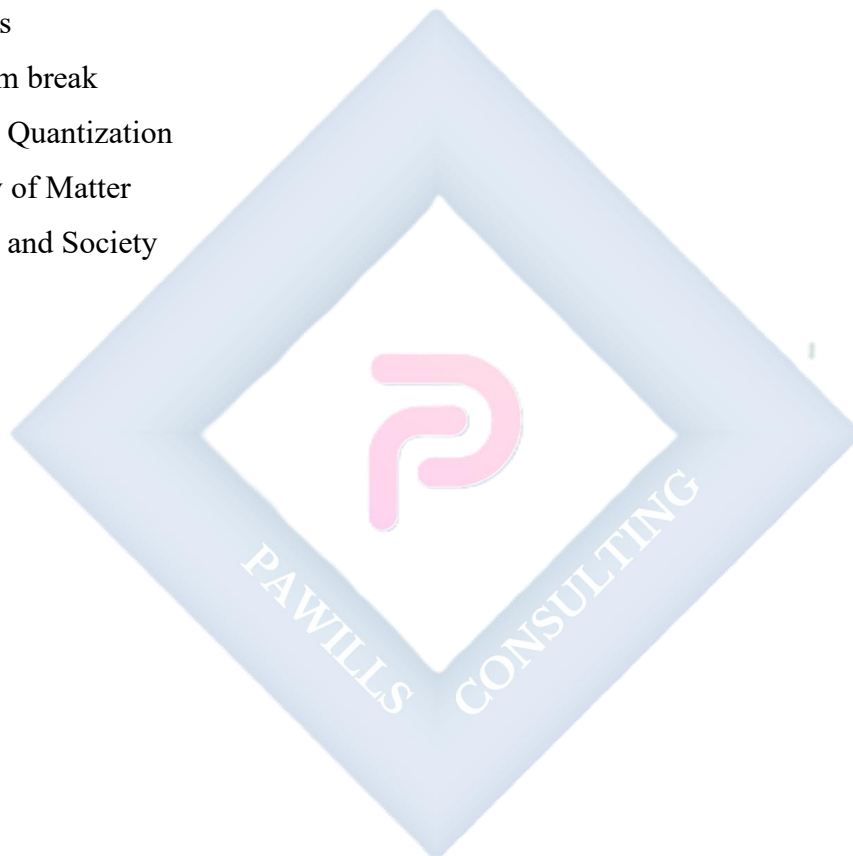
- (a) With a diagram, explain the structure of a flat-plate solar collector.
- (b) Research and describe how solar collectors are used in large-scale applications like solar water heating systems in hotels.
- (c) List five safety or maintenance practices to ensure long life of a solar thermal system.



SS2 PHYSICS LESSON NOTES

THIRD TERM

1. Electrical Continuity Testing
2. Electroplating
3. Application of Electromagnetic Field
4. Models of the Atom
5. Midterm examination
6. Nucleus
7. Midterm break
8. Energy Quantization
9. Duality of Matter
10. Energy and Society



WEEK 1 - ELECTRICAL CONTINUITY TESTING

DEFINITION OF ELECTRICAL CONTINUITY

Electrical continuity refers to the **uninterrupted flow of electric current** through a conductive path or circuit.

- A conductor or component is said to be **continuous** if it forms a **complete, unbroken path** for electrons to flow from one terminal to another.
- A **discontinuity** (such as a cut wire, broken fuse, or faulty switch) interrupts the circuit, resulting in **zero or no current flow**.

Continuity Testing: This is a diagnostic procedure used to verify the presence or absence of a complete path for current flow within an electrical or electronic circuit.

ELECTRICAL PRINCIPLE BEHIND CONTINUITY TESTING

The concept is based on **Ohm's Law**:

$$V = IR$$

- In continuity testing, the goal is to determine if **resistance RRR** is low enough (close to **0 ohms**) to allow current to pass.
- A **low or zero resistance** implies continuity (closed path), while a **very high or infinite resistance** implies an open circuit (discontinuity).

In continuity mode, a multimeter sends a **small voltage/current** across the circuit and detects whether a return path exists.

TOOLS USED FOR CONTINUITY TESTING

(i) Digital Multimeter:

- Most reliable instrument for testing.
- In **continuity mode**, it produces a beep if continuity exists.
- Can also be used to measure **resistance** to determine continuity indirectly.

(ii) Analog Multimeter:

- Uses a needle to display low resistance values.
- Absence of deflection means no continuity.

(iii) Continuity Tester:

- A simple handheld tool with a built-in battery and bulb or buzzer.
- Lights up or buzzes when the circuit is complete.

(iv) Test Lamp:

- Basic tool made from a bulb, battery, and two leads.
- Lights up if current flows through the path (i.e., continuity exists).

PROCEDURE FOR CONTINUITY TESTING (USING MULTIMETER)

Safety First:

- Always **turn off the power supply** to the circuit/device before testing.

Steps:

1. Set the multimeter to continuity mode (or lowest ohm range if not available).
2. Insert the probes into the multimeter (black to COM, red to Ω port).
3. Touch the two probes together to confirm the meter is working (it should beep).
4. Touch the probes to both ends of the component or wire being tested.
5. Observe the reading or listen for the beep.

Interpretation:

Reading/Signal	Meaning
0 to $<1\Omega$ / beeping sound	Continuity (closed path)
$>1000\Omega$ / No beep	No continuity (open path)
“OL” or “ ∞ ”	Infinite resistance (broken)

COMPONENTS THAT CAN BE TESTED FOR CONTINUITY

- Electrical wires (live, neutral, earth)
- Fuses (to check if blown)
- Switches (open/close contact verification)
- Circuit breakers
- Conductive tracks on PCB
- Coils in motors or transformers
- Connectors or terminals

COMMON FAULTS IDENTIFIED THROUGH CONTINUITY TESTING

Fault Type	Description
Open Circuit	Disconnected wire/component; no current flow
Short Circuit	Unintended low-resistance path
Blown Fuse	Melted conductor inside fuse interrupts path
Loose Connector	Weak contact causing intermittent breaks
Broken Trace	Damaged path in PCB or wire internally

APPLICATIONS OF CONTINUITY TESTING

1. **Electrical Installation:** Verifying wiring connections in homes/offices.
2. **Automotive Repair:** Checking wiring harnesses, sensors, and ignition paths.
3. **Electronics Production:** Quality control in circuit boards.
4. **Maintenance:** Locating faults in old electrical systems.
5. **Industrial Control Systems:** Ensuring proper signal flow in control panels.
6. **Aerospace & Rail Systems:** Safety-critical integrity verification.
7. **Instrument Calibration:** Confirming connections before power-up.

SAFETY PRECAUTIONS

1. Never perform continuity testing on a **powered or live circuit**.
2. Ensure **capacitors are discharged** before testing.
3. Use **correct range settings** on the multimeter.
4. **Inspect probes** for damage before use.

5. Handle tools with **dry hands and rubber soles**.
6. Always **label tested wires** to avoid confusion.
7. Keep **flammables away** when testing.

CALCULATIONS AND PHYSICS APPLICATIONS

While continuity testing is primarily qualitative (pass/fail), it may involve resistance measurements and concepts from **Ohm's Law**.

Example Problem:

A resistor of resistance $1.5\ \Omega$ is tested using a continuity function. If the multimeter shows a value of 1.5, does this indicate continuity?

Answer: Yes. The resistance is low, so continuity exists.

EVALUATION

Objective Questions (WAEC/NECO Style):

1. Continuity testing is used to:
 - A. Measure power
 - B. Detect open circuits ✓
 - C. Measure voltage
 - D. Generate electric current
2. A good conductor when tested for continuity shows:
 - A. High resistance
 - B. No reading
 - C. A beep sound ✓
 - D. Spark

Theory Questions:

1. Define continuity testing and explain how it works.
2. Describe how a test lamp can be used to detect a broken wire.
3. Differentiate between continuity test and resistance measurement.
4. List five components that can be tested for continuity and the interpretation of results.

5. A student tests a switch using a multimeter and hears no sound. What could this imply?
Suggest a remedy.

ASSIGNMENT

- (a) Sketch a circuit diagram showing a continuity tester connected to a wire and a battery.
- (b) Explain the step-by-step procedure for testing continuity in a car's headlamp wiring system.
- (c) In what ways is continuity testing useful in electronics manufacturing industries?



WEEK 2 - ELECTROPLATING

DEFINITION OF ELECTROPLATING

Electroplating is the process of depositing a thin layer of one metal onto the surface of another metal object using **electric current** and an **electrolyte solution**. It is a type of **electrolysis** used mainly for **coating, protection, decoration, or improving conductivity** of objects.

BASIC PRINCIPLE OF ELECTROPLATING

- Based on **electrolysis**, where **direct current (DC)** passes through an electrolyte solution containing metal ions.
- **The metal to be deposited** is connected as the **anode**, and the **object to be coated** is the **cathode**.
- When electricity flows, **positive metal ions from the electrolyte** are reduced at the cathode and form a **coating**.
- Meanwhile, **anode dissolves** to replenish metal ions in solution (if it's made of the plating metal).

COMPONENTS OF AN ELECTROPLATING SYSTEM

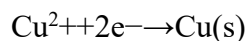
Component	Description
Power Supply	Provides direct current (DC) to drive the reaction
Cathode (-)	The object to be plated (e.g., steel spoon)
Anode (+)	Metal used for plating (e.g., copper, nickel)
Electrolyte	A solution containing ions of the plating metal (e.g., CuSO_4)
Connecting wires	To connect the electrodes to the power supply

ELECTROCHEMICAL REACTIONS IN ELECTROPLATING

Example: Electroplating a spoon with copper

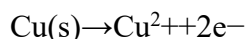
- **Electrolyte:** Copper(II) sulfate solution (CuSO_4)
- **Anode:** Copper plate
- **Cathode:** Iron or steel spoon

At Cathode (-):



Copper ions are reduced and deposited as metallic copper on the spoon.

At Anode (+):



Copper from the anode dissolves to replace ions in the solution.

The **thickness of the deposited metal** depends on **current, time, concentration, and temperature**.

GENERAL CONDITIONS FOR EFFECTIVE ELECTROPLATING

- Clean surface of the object to be plated
- Constant DC current supply
- Correct concentration of electrolyte
- Adequate temperature (moderately warm is ideal)
- Proper electrode placement
- Duration long enough for desired thickness

COMMON METALS USED FOR ELECTROPLATING

Plating Metal	Reason
Nickel	Corrosion resistance and aesthetic shine
Chrome	Hardness, corrosion resistance
Gold/Silver	Decorative, conductivity (electronics)
Zinc	Protects steel from rust (galvanization)
Copper	Base layer for other plating or aesthetics

APPLICATIONS OF ELECTROPLATING

1. Jewelry coating with gold or silver
2. Automobile parts (bumpers, handles)
3. Cutlery and kitchenware

4. Electrical connectors and circuit boards
5. Coins and medals
6. Protective coating on tools and machines
7. Galvanization (zinc-plating of iron to prevent rust)

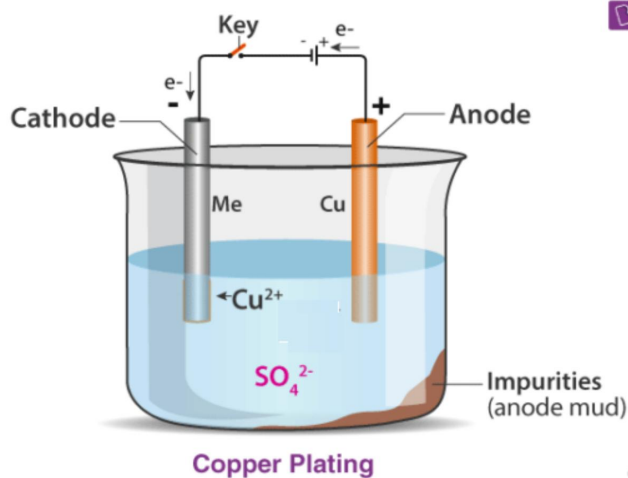
ADVANTAGES OF ELECTROPLATING

1. Enhances appearance (shiny, decorative)
2. Prevents corrosion and rust
3. Improves electrical conductivity
4. Increases surface hardness and durability
5. Reduces friction in machinery parts
6. Makes cheaper metals appear valuable (e.g., gold-plated items)

DISADVANTAGES / LIMITATIONS OF ELECTROPLATING

1. **Costly** if precious metals (e.g., gold, silver) are used
2. **Toxic chemicals** used can harm environment if not properly managed
3. Coated layer may **wear off** over time
4. Requires **controlled conditions** for quality results
5. Improper cleaning of surfaces leads to **poor adhesion**

Electroplating Setup Diagram:



SAFETY PRECAUTIONS IN ELECTROPLATING

1. Use **gloves and goggles** to handle chemicals.
2. Work in a **well-ventilated** area.
3. Avoid **direct skin contact** with electrolyte.
4. Do not allow electrodes to **touch** inside the beaker.
5. Wash hands and materials after the experiment.
6. Store chemicals **safely and label properly**.
7. **Dispose of used solutions** responsibly (do not pour into drains).

EVALUATION

Objective Questions (WAEC/NECO Style):

1. In electroplating, the object to be plated is connected to the:
 - A. Positive terminal
 - B. Negative terminal ✓
 - C. Earth
 - D. Neutral wire
2. Electroplating involves:
 - A. Magnetic induction
 - B. Chemical bonding
 - C. Electrolysis ✓
 - D. Radiation transfer

Theory Questions:

1. Define electroplating and explain its working principle.
2. Draw and label a simple electroplating setup.
3. State four uses of electroplating in the industry.
4. What are the reactions at the anode and cathode during copper electroplating?
5. List three factors that influence the quality of electroplating.
6. State two advantages and two disadvantages of electroplating.

ASSIGNMENT

- (a) Research and describe the industrial process of **electroplating with chromium** for car parts.
- (b) Explain how electroplating is used in preventing rust in steel bridges.
- (c) Describe an experiment to electroplate an iron nail with zinc. Include materials and procedure.



WEEK 3 - APPLICATION OF ELECTROMAGNETIC FIELD

DEFINITION OF ELECTROMAGNETIC FIELD (EMF)

An **electromagnetic field (EMF)** is a physical field produced by electrically charged objects. It affects the behavior of charged particles in its vicinity. The field consists of two components:

- **Electric field (E-field):** Produced by stationary charges. It exerts force on other electric charges in the field.
- **Magnetic field (B-field):** Produced by moving charges (current). It exerts force on other moving charges.

The electromagnetic field is governed by **Maxwell's equations**, which describe how electric and magnetic fields are generated and altered by each other and by charges and currents.

Interaction Between Electric and Magnetic Fields

Electric and magnetic fields are inherently linked. When an electric charge moves (i.e., a current flows), it produces a magnetic field. Similarly, a changing magnetic field induces an electric field (as described by **Faraday's Law of Electromagnetic Induction**).

This interaction forms the basis of electromagnetic waves (e.g., radio waves, light). In an electromagnetic wave, the electric and magnetic fields oscillate perpendicular to each other and to the direction of wave propagation.

LAWS OF ELECTROMAGNETIC INDUCTION:

Faradays law states that whenever there is a change in the magnetic lines of force (e.m.f) is induced, the strength which is proportional to the rate of change of the flux linked with the circuit.

Lenz's law states that the induced e.m.f is in such a direction as to oppose the motion or change producing it.

COMMON APPLICATIONS OF ELECTROMAGNETIC FIELDS IN EVERYDAY LIFE

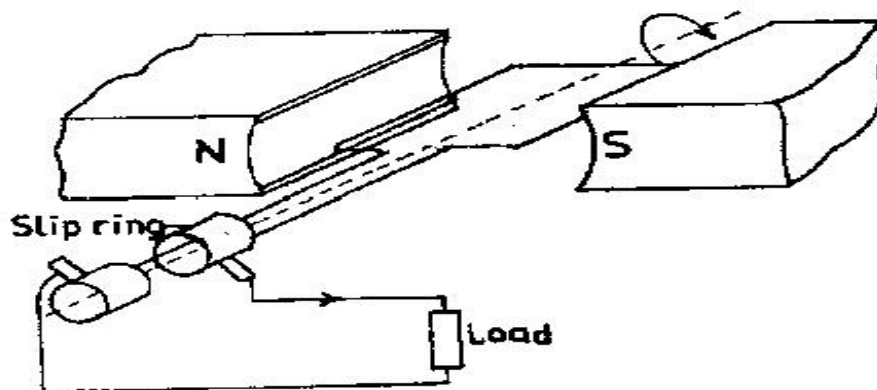
Electromagnetic fields are pervasive in modern society. Some common applications include:

- **Electric Power Transmission:** Transformers and high-voltage lines rely on alternating EMFs to transmit electricity.
- **Household Appliances:** Devices like microwaves, toasters, and induction cookers use EMFs for heating or operation.
- **Communication Systems:** Radio, television, mobile phones, and Wi-Fi depend on electromagnetic wave propagation.
- **Medical Equipment:** MRI scanners and diathermy machines use controlled EMFs for diagnostic and therapeutic purposes.
- **Security Systems:** Metal detectors, RFID tags, and airport scanners operate using electromagnetic principles.

WORKING PRINCIPLES OF ELECTROMAGNETIC DEVICES

Eddy currents are loops of electrical current induced between conductors by changing magnetic field in the conductor. Eddy current flows in closed loops within conductors in planes perpendicular to the magnetic field. Eddy flows in the form of Eddies, in such a direction so as to oppose the motion of the conductor in the field.

Mutual Inductance: This is the flow of induced current or voltage in a coil due to an alternating current in a neighbouring coil.



The diagram above illustrates an a.c. generator. When the coil is rotated, e.m.f. is reduced in the coil.

The purpose of the slip rings is to obtain an alternating voltage.

Electric Motors:

Convert electrical energy into mechanical energy. When current flows through a coil in a magnetic field, a force (Lorentz force) acts on the coil, causing it to rotate (based on **Fleming's Left-Hand Rule**).

1. Electric Generators:

Convert mechanical energy into electrical energy using **electromagnetic induction**.

Movement of a conductor (coil) in a magnetic field induces current (based on **Fleming's Right-Hand Rule**).

2. Transformers:

Transfer electrical energy between circuits via mutual induction. They consist of primary and secondary coils wound around a magnetic core. A changing current in the primary induces a voltage in the secondary.

3. Loudspeakers:

Convert electrical signals into sound. An alternating current in a coil causes vibration in a diaphragm attached to the coil, producing sound waves. This relies on the interaction between the coil and a permanent magnet.

4. Electromagnetic Relays:

Electrically operated switches. A small current activates an electromagnet, which moves a switch to allow a larger current to flow through another circuit.

TECHNOLOGICAL APPLICATIONS OF ELECTROMAGNETIC FIELDS

1. Magnetic Resonance Imaging (MRI):

A diagnostic tool in medicine that uses strong magnetic fields and radio waves to produce detailed images of internal body structures. The magnetic field aligns the hydrogen nuclei in the body, and the radio waves disturb this alignment. The return to equilibrium emits signals used to construct images.

2. **Induction Heating:**

Uses electromagnetic induction to heat electrically conductive materials (e.g., metals). An alternating current produces eddy currents in the material, generating heat due to electrical resistance. It is widely used in metallurgy and cookware (e.g., induction stoves).

3. **Wireless Power Transfer:**

Transfers electrical energy without wires using oscillating magnetic fields (resonant inductive coupling). Common in wireless charging of phones, electric toothbrushes, and even electric vehicles.

SIMPLE DEVICES USING ELECTROMAGNETIC FIELDS

(i) Electromagnetic Relay

- A switch controlled by an electromagnet.
- When current flows, the magnet pulls a contact to close/open a circuit.

(ii) Electric Bell

- Works when an EM field pulls an armature, breaks contact, and allows repeated ringing action.

(iii) Loudspeaker

- Coil in magnetic field vibrates diaphragm to produce sound in response to EM field changes.

ADVANTAGES OF USING ELECTROMAGNETIC FIELDS

- Enable **contactless control** of devices
- Allow **wireless communication**
- Efficient **energy conversion** (motors/generators)
- Used in **high-speed, non-contact transport systems**
- **Medical imaging** without surgical intrusion

DISADVANTAGES / RISKS OF EM FIELD EXPOSURE

- **Strong EM fields** can interfere with pacemakers or sensitive electronics
- Prolonged exposure to **high-intensity EM radiation** (e.g., X-rays) can be harmful
- **Electromagnetic interference (EMI)** in critical systems (e.g., aircraft)

- Potential **health risks** from long-term exposure to high-frequency fields (e.g., phone towers)

Factors Which Determine The Magnitude Of An Induced *Emf* In A Coil Are:

- Speed of rotation of the coil
- Cross section area of the coil
- Number of turns in the coil.

EVALUATION (WAEC/NECO PAST QUESTIONS)

1. (WAEC June 2007 Q4)

Explain briefly the principle of operation of an electric motor and state two factors that determine its direction of rotation.

2. (WAEC June 2015 Q6)

(a) Define electromagnetic induction.

(b) State two applications of electromagnetic induction in everyday life.

3. (NECO 2019 Q8)

Describe how a transformer works and mention two devices that use transformers.

4. (WAEC June 2014 Q9a)

Mention two appliances that use electromagnetic fields and describe their working principles.

ASSIGNMENT

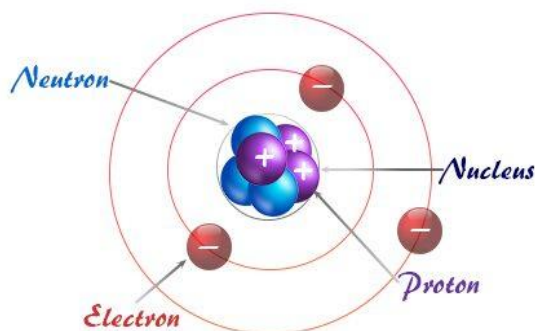
1. Define electromagnetic field and describe its two components.
2. Explain how electromagnetic fields are utilized in:
 - a. MRI
 - b. Induction heating
 - c. Wireless charging
3. With the aid of diagrams, explain the principle of operation of either a generator or an electric motor.
4. Discuss two societal concerns related to electromagnetic field exposure and how they are mitigated.
5. Explain eddy currents.

WEEK 4 - MODELS OF THE ATOM

HISTORICAL BACKGROUND OF THE ATOMIC MODEL

Ancient Ideas (Democritus, ~400 BC)

- Proposed the idea that matter is composed of tiny indivisible particles called "atomos" (meaning uncuttable).
- Had no experimental evidence—purely philosophical.



Atomic Structure of Lithium

Priyamstudycentre.com

DEVELOPMENT OF MODERN ATOMIC MODELS

(1) Dalton's Atomic Model (1803)

Scientist: John Dalton

Nature of the Model: Indivisible solid sphere

Postulates:

1. All matter is composed of indivisible atoms.
2. Atoms of the same element are identical in mass and properties.
3. Atoms combine in simple whole-number ratios to form compounds.
4. Chemical reactions involve the rearrangement of atoms.

Strengths:

- First scientific model supported by experimentation.
- Explains conservation of mass and definite proportions.

Limitations:

- Did not include subatomic particles (electrons, protons, neutrons).

- Failed to explain isotopes and chemical bonding.

(2) Thomson's Model – “Plum Pudding Model” (1897)

Scientist: J.J. Thomson

Experiment: Cathode Ray Tube

Discovery: Electron

Model Description:

- Atom is a positively charged sphere with negatively charged electrons embedded like “raisins in a pudding”.

Key Observations from CRT Experiment:

- Cathode rays (electrons) are negatively charged particles.
- They are deflected by electric and magnetic fields.

Strengths:

- Introduced subatomic particles.
- Explained electrical conductivity.

Limitations:

- No central nucleus.
- Could not explain atomic stability.
- No explanation for spectral lines.

(3) Rutherford's Nuclear Model (1911)

Scientist: Ernest Rutherford

Experiment: Alpha Scattering (Gold Foil) Experiment

Setup: Alpha particles bombarded a thin gold foil.

Observations:

1. Most alpha particles passed through.
2. A few were deflected.
3. Some bounced back.

Conclusions:

- Atom is mostly empty space.

- The atom has a small, dense, positively charged nucleus.
- Electrons orbit the nucleus.

Strengths:

- Introduced the nuclear structure of the atom.
- Explained scattering of alpha particles.

Limitations:

- Could not explain why negatively charged electrons don't spiral into the nucleus.
- Could not explain atomic line spectra.

(4) Bohr's Model (1913)

Scientist: Niels Bohr

Improvement on Rutherford's model using Quantum Theory

Postulates:

1. Electrons revolve in fixed circular orbits (energy levels or shells).
2. Electrons do not radiate energy while in these orbits.
3. Electrons emit/absorb energy when transitioning between orbits:

Energy Level Formula:

$$E_n = -\frac{13.6}{n^2} \text{eV}$$

Where n = principal quantum number

Successes:

- Explained hydrogen atom's discrete spectral lines.
- Introduced quantization of angular momentum:

$$mvr = \frac{nh}{2\pi}$$

Limitations:

- Only accurately explained hydrogen and single-electron systems.
- Could not explain Zeeman and Stark effects.
- Ignored wave-particle duality.

(5) Quantum Mechanical Model / Schrödinger Model (1926–Present)

Scientists: Schrödinger, Heisenberg, de Broglie, Dirac

Key Ideas:

- Electrons do not move in fixed orbits, but in **probability clouds** or **orbitals**.
- Describes electron behavior using complex mathematical wave functions:
 $\psi(x,t)$
- Heisenberg Uncertainty Principle:**
(It is impossible to know both position and momentum of an electron simultaneously.)
- de Broglie Wavelength:**

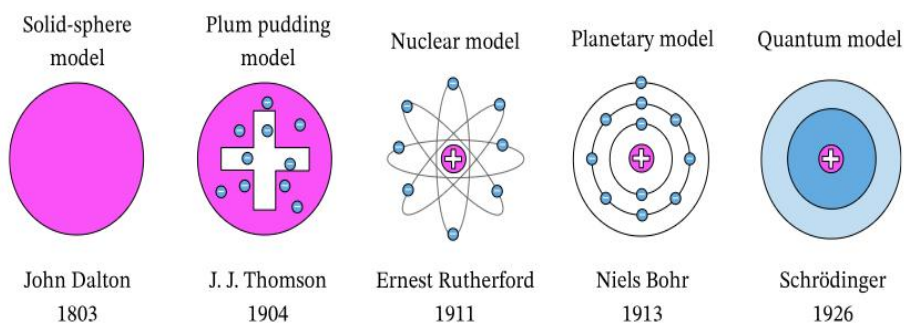
$$\lambda = \frac{mv}{h}$$

Strengths:

- Describes all atoms (including complex multi-electron atoms).
- Foundation for quantum chemistry and particle physics.

Limitations:

- Highly mathematical.



- Difficult to visualize.

COMPARISON TABLE OF ATOMIC MODELS

Model	Scientist	Description	Strengths	Limitations
Dalton	John Dalton	Solid indivisible atom	Explains conservation of mass	No subatomic structure
Thomson	J.J. Thomson	Electrons in positive	Introduced electrons	No nucleus

		sphere		
Rutherford	Ernest Rutherford	Dense nucleus with orbiting electrons	Discovery of nucleus	Could not explain electron stability
Bohr	Niels Bohr	Quantized circular orbits	Explained hydrogen spectra	Inaccurate for complex atoms
Quantum	Schrödinger et al	Electron clouds and probability orbitals	Most accurate, includes wave nature	Highly mathematical

APPLICATIONS OF ATOMIC MODELS

1. **Chemistry:** Understanding molecular bonding and reactions.
2. **Spectroscopy:** Explanation of atomic spectra, emission and absorption lines.
3. **Electronics:** Semiconductor design and operation.
4. **Medical Imaging:** MRI, PET scans rely on atomic structure.
5. **Nuclear Physics:** Nuclear fission, fusion, atomic energy.
6. **Quantum Computing:** Based on manipulation of electron states.
7. **Laser Technology:** Based on electron energy level transitions.

EVALUATION

Objective Questions (Multiple Choice)

1. Which model introduced the concept of quantized electron orbits?
 - A. Dalton's Model
 - B. Bohr's Model ✓
 - C. Rutherford's Model
 - D. Thomson's Model
2. The discovery of the electron led to which atomic model?
 - A. Plum pudding ✓
 - B. Nuclear model
 - C. Planetary model
 - D. Quantum model

3. Which model introduced the idea of orbitals and probability clouds?

- A. Bohr's
- B. Dalton's
- C. Quantum Mechanical ✓
- D. Rutherford's

Theory Questions

1. List and explain four postulates of Dalton's atomic theory.
2. Describe Rutherford's gold foil experiment and state two of its conclusions.
3. Outline three limitations of Bohr's atomic model.
4. Compare Bohr's model with Schrödinger's quantum mechanical model.
5. Explain how atomic models helped in the development of X-ray and spectroscopy.

ASSIGNMENT

(a) Draw and label diagrams for:

- Bohr's hydrogen atom
- Schrödinger's s and p orbital
- Rutherford's gold foil setup

(b) State how the concept of electron orbitals has influenced modern computer chip technology.

(c) Research the contribution of **Heisenberg** and how uncertainty plays a role in quantum mechanics.

WEEK 6 - NUCLEUS

DEFINITION OF NUCLEUS

The *nucleus* is the small, dense center of an atom where most of the atom's mass is concentrated. It was discovered through the gold foil experiment conducted by Ernest Rutherford in 1911. The nucleus is positively charged and occupies only a tiny fraction of the volume of the atom but contains nearly all its mass.

CONSTITUENTS OF THE NUCLEUS

The nucleus is made up of two fundamental subatomic particles:

1. Protons

- Positively charged particles.
- Each proton carries a charge of $+1.6 \times 10^{-19}$ C.
- The number of protons determines the *element* and is called the **proton number** or **atomic number (Z)**.

2. Neutrons

- Electrically neutral particles.
- Similar mass to protons but no charge.
- Neutrons contribute to the *mass* of the atom and influence nuclear stability.

Together, protons and neutrons are called **nucleons**.

NUCLEON NUMBER, PROTON NUMBER, ISOTOPES, AND NUCLIDES

1. Nucleon Number (Mass Number, A):

- Total number of protons and neutrons in the nucleus.
- $A = Z + N$

Where:

- A = Nucleon number
- Z = Number of protons (proton number)
- N = Number of neutrons

2. Proton Number (Z):

- Number of protons in the nucleus.
- Defines the chemical identity of an atom (e.g., hydrogen has $Z = 1$).

3. Isotopes:

- Atoms of the same element (same proton number) but different nucleon numbers.
- Example:
 - Hydrogen isotopes:
 - Protium (^1H), Deuterium (^2H), Tritium (^3H)

4. Nuclides:

- A nuclide is a specific isotope of an element characterized by its proton and neutron numbers.
- Represented as:
 ${}^Z_{\text{A}}\text{X}$
Example: ${}^{14}_{6}\text{C}$ represents a carbon nuclide with 6 protons and 8 neutrons.

NUCLEAR PROPERTIES

1. Mass Defect (Δm):

- The difference between the sum of the masses of individual nucleons and the actual mass of the nucleus.
- Indicates that some mass has been converted into energy that binds the nucleons together.

2. Binding Energy (BE):

- The energy required to split a nucleus into its individual protons and neutrons.
- It reflects the stability of the nucleus.
- Calculated using Einstein's mass-energy equivalence:

$$E = \Delta m \cdot c^2$$

Where:

- E = Binding energy (in joules or MeV)
- Δm = Mass defect (in kg or u)
- c = Speed of light (3×10^8 m/s)

3. Nuclear Stability:

- Stable nuclei have a balanced ratio of protons to neutrons.

- Too many or too few neutrons can lead to radioactive decay.

4. **'half life':** The half life of a substance is defined as the time taken for half of the atoms in any given substance to decay

$$T = \frac{0.692}{\lambda}$$

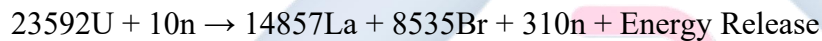
T = half life

λ = decay constant

NUCLEAR FISSION AND FUSION

Nuclear Fission: This is a process in which a nuclear bombardment of the nucleus of a heavy nuclide by particles such as a neutrons, resulting in splitting of the nucleus into two smaller nuclei with the release of large amount of energy.

Example:



- A heavy nucleus (e.g., uranium-235) splits into two lighter nuclei with the release of a large amount of energy and free neutrons.
- Used in nuclear reactors and atomic bombs.
- Chain reaction can occur if released neutrons initiate further fissions.

Nuclear Fusion:

- Two light nuclei (e.g., hydrogen isotopes) combine to form a heavier nucleus.
- Releases even more energy than fission.
- Occurs naturally in stars, including the sun.
- Requires extremely high temperature and pressure.

COMPONENT OF THE NUCLEAR REACTOR

- nuclear fuels
- nuclear moderator
- control rods
- Reactor coolants

- The turbine
- Cooling towers.

FUNCTION OF NUCLEAR FUELS

- The fuel help to supply the needed energy would power the nuclear reactor .

Function nuclear moderators

- They are used to slow down fast moving neutron which can cause further fission. Examples include graphite

function of controls rods

- they are used to absorb neutron which can lead to further fission. Examples are boron, cadmium e.t.c

Function of reactor coolants.

They are used to carry th heat produced by fission to external boiler and turbine where it is transformed into electricity, examples are liquid sodium, lead e.t.c

Function of turbine

- it transfer to heat from coolant to electricity

Function of cooling towers

- They are used to dump the excess heat that cannot be converted to energy due to the laws of thermodynamics.

USES AND HAZARDS OF NUCLEAR ENERGY

Applications:

1. Medicine:

- Radiation therapy for cancer treatment.
- Radioisotopes used in diagnostic imaging (e.g., iodine-131 for thyroid imaging).

2. Industry:

- Non-destructive testing of materials using radioactive tracers.
- Sterilization of medical equipment.

3. Power Generation:

- Nuclear reactors generate electricity using controlled fission.

- Example: Pressurized Water Reactor (PWR), Boiling Water Reactor (BWR)

Hazards:

- Exposure to high doses of radiation can cause cancer, genetic mutations, and acute radiation sickness.
- Radioactive waste disposal poses long-term environmental and health risks.
- Nuclear accidents (e.g., Chernobyl, Fukushima) can have catastrophic consequences.

Einstein's Mass-Energy Equivalence and Binding Energy Calculation

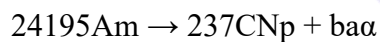
The equation:

$$E = \Delta m \cdot c^2$$

is used to calculate binding energy when the mass defect is known. Binding energy per nucleon is an important measure of nuclear stability.

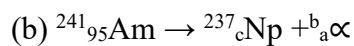
Example 2:

A radioactive isotope of Americium (Am - 241) decays into nucleus of Neptunium (Np - 237) and an alpha (α) particle as shown in the nuclear equation below.

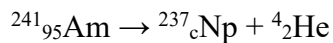


Determine the values of a, b and c.

Solution:



Recall that ${}^b_a\alpha = {}^4_2\text{He}$



$$(ii) 95 = c + 2; c = 93$$

$$(i) \text{Number of neutrons} = 241 - 95 = 146$$

Example 2:

If a nucleus has a mass defect $\Delta m = 0.0025 \text{ u}$, convert to kg:

Solution

$$\Delta m = 0.0025 \times 1.66 \times 10^{-27} \text{ kg} = 4.15 \times 10^{-30} \text{ kg}$$

$$E = 4.15 \times 10^{-30} \times (3 \times 10^8)^2 = 3.735 \times 10^{-13}$$

Alternatively, use $1 \text{ u} = 931 \text{ MeV}$ for direct energy conversion:

$$E = 0.0025 \times 931 = 2.33 \text{ MeV}$$

Example 3:

In a nuclear reaction, the mass defect is $2.0 \times 10^{-6} \text{ g}$. Calculate the energy released, given that the velocity of light is $3.0 \times 10^8 \text{ ms}^{-1}$.

Solution:

$$E = \Delta m c^2 = 2.0 \times 10^{-6} \text{ g} = 2.0 \times 10^{-9} \text{ kg}; c = 3 \times 10^8 \text{ ms}^{-1}$$

$$E = \Delta m c^2 = 2.0 \times 10^{-9} \times (3 \times 10^8)^2 = 2.0 \times 10^{-9} \times 9 \times 10^{16} = 18 \times 10^7 \text{ J} = 1.8 \times 10^8 \text{ J}$$

Evaluation (WAEC/NECO Past Questions)

1. **(WAEC 2019 Q6):** Explain the difference between mass number and atomic number of an element.
2. **(NECO 2021 Q9):** State two differences between nuclear fission and fusion.
3. **(WAEC 2022 Q3):** A helium nucleus has a mass defect of 0.0304 u . Calculate its binding energy. ($1 \text{ u} = 931 \text{ MeV}$)
4. **(WAEC 2018 Q10):** Give one medical and one industrial application of nuclear energy.
5. **(NECO 2020 Q4):** What is an isotope? Give two examples of isotopes of hydrogen.

Assignment

1. Define the following terms: nuclide, isotope, binding energy.
2. Differentiate between nuclear fission and nuclear fusion with one example each.
3. Calculate the binding energy in MeV of a lithium nucleus that has a mass defect of 0.035 u .

4. Mention two advantages and two disadvantages of using nuclear energy in power generation.
5. Give one example each of the use of nuclear radiation in:
 - Medicine
 - Industry
 - Agriculture



WEEK 8 - ENERGY QUANTIZATION

INTRODUCTION

In classical physics, energy was thought to vary continuously. However, early 20th-century discoveries in atomic and quantum physics revealed that energy, especially at the atomic and subatomic levels, exists in **discrete packets**, not continuous forms. This led to the concept of **energy quantization**, a fundamental idea in quantum physics that radically transformed our understanding of atomic structure and electromagnetic radiation.

DEFINITION OF ENERGY QUANTIZATION

Energy quantization refers to the concept that energy levels of electrons in an atom are not continuous, but rather exist in **discrete (quantized) energy states**. This means that an electron in an atom can only possess certain specific energy values and not just any arbitrary amount.

These energy states are associated with **orbitals or energy levels (shells)** around the nucleus. An electron must **absorb** or **emit** a **fixed amount of energy**, called a **quantum**, to move from one energy level to another.

DISCRETE ENERGY LEVELS IN ATOMS

Atoms have a set of **fixed energy levels**, denoted by quantum numbers ($n = 1, 2, 3, \dots$). These energy levels are like rungs of a ladder — an electron can "stand" on one rung or another, but **cannot exist between rungs**.

- **Ground state:** The lowest energy level ($n = 1$), where the electron is most stable.
- **Excited state:** Any energy level higher than the ground state ($n = 2, 3, \dots$), where the electron has gained energy.

Each level corresponds to a **specific amount of energy**, and the difference between any two levels represents the energy an electron must absorb or emit during a transition.

ELECTRON TRANSITIONS AND QUANTA OF ENERGY

Electrons can move between energy levels by either:

- **Absorbing energy:** Moving to a higher energy level (excitation).
- **Emitting energy:** Falling to a lower energy level (de-excitation).

The energy exchanged during these transitions is in the form of **electromagnetic radiation** (like light), and it is always a **discrete quantum of energy**, not a continuous flow.

This energy is given by:

$$\Delta E = E_{\text{higher}} - E_{\text{lower}} = hf$$

Where:

- ΔE = energy absorbed/emitted,
- h = Planck's constant (6.626×10^{-34} J
- f = frequency of the absorbed/emitted radiation.

PLANCK'S HYPOTHESIS AND DERIVATION OF $E = hf$

In 1900, **Max Planck** introduced the idea that electromagnetic energy is quantized. His hypothesis stated that:

"Energy is emitted or absorbed in discrete units or quanta, and the energy E of each quantum is directly proportional to the frequency f of radiation."

Mathematically, this is expressed as:

$$E = hf$$

Where:

- E = energy of a single quantum of radiation (in joules),
- h = Planck's constant,

- f = frequency of radiation (in hertz, Hz).

This formula marked the beginning of **quantum theory**, breaking away from classical physics, which treated energy as continuous.

THE PHOTOELECTRIC EFFECT

The **photoelectric effect** is a phenomenon where **electrons are ejected from the surface of a metal** when light of sufficient frequency shines on it.

Experimental Observations:

1. No electrons are emitted unless the light's frequency is above a **threshold frequency** (f_0), regardless of intensity.
2. The kinetic energy of emitted electrons depends on the **frequency** of the light, not its intensity.
3. The number of electrons emitted depends on the **intensity** (brightness) of the light, if frequency is above threshold.
4. The emission of electrons is **instantaneous** when the threshold frequency is reached.

These observations could **not be explained** by classical wave theory, but were successfully explained by Einstein using Planck's quantum hypothesis.

Einstein proposed that light is made up of particles called **photons**, each carrying energy $E=hf$. When a photon strikes an electron, it transfers energy:

$$hf = \phi + \frac{1}{2}mv^2$$

Where:

- ϕ = work function (minimum energy needed to free an electron),
- $\frac{1}{2}mv^2$ = kinetic energy of the emitted electron.

This confirmed that energy is quantized, further validating Planck's idea.

QUANTIZATION AND SPECTRAL LINES

When electrons in an atom move between energy levels, they **emit or absorb specific amounts of energy**. These energy changes correspond to particular **frequencies or wavelengths** of light.

Each element emits light at **specific frequencies**, producing a **line spectrum** (as opposed to a continuous spectrum). This is known as an **atomic emission or absorption spectrum**.

For example:

- **Hydrogen atom** emits visible light at specific lines in the **Balmer series**, corresponding to electron transitions to the $n=2$ level.

These spectral lines serve as **fingerprints of elements**, used in chemical analysis and astronomy.

SIGNIFICANCE OF ENERGY QUANTIZATION

- It explains **stability of atoms** — electrons cannot spiral into the nucleus as they can only occupy fixed levels.
- It accounts for **discrete spectral lines** in emission and absorption spectra.
- It supports the understanding of **photoelectric effect, blackbody radiation, and atomic models** like Bohr's model.
- It forms the basis for **quantum mechanics**, modern electronics, lasers, and semiconductors.

EVALUATION (WAEC/NECO PAST QUESTIONS)

1. **(WAEC 2017, Q7)**: The energy of a photon is 3.0×10^{-19} J. Calculate the frequency of the radiation. ($h = 6.63 \times 10^{-34}$ Js)
2. **(NECO 2015, Q5)**: Explain why light of frequency below a certain value does not cause photoemission from a metal surface.
3. **(WAEC 2016, Q4)**: Describe the photoelectric effect and explain how it supports the quantum theory of light.
4. **(WAEC 2014, Q3b)**: Define the term energy quantization and explain its role in the emission of line spectra.

5. (NECO 2020, Q6): Derive the relationship $E=hf$ and state its significance.

ASSIGNMENT

1. Define the following terms:
 - a) Energy quantization
 - b) Photon
 - c) Threshold frequency
 - d) Work function
2. An electron in an atom absorbs 4.14×10^{-19} J of energy. Calculate the frequency of the absorbed radiation. ($h = 6.63 \times 10^{-34}$ Js)
3. Discuss how the photoelectric effect cannot be explained by classical physics but supports quantum theory.
4. Explain why the hydrogen atom emits light of specific wavelengths.
5. Distinguish between continuous spectrum and line spectrum with examples.



WEEK 9 - DUALITY OF MATTER

INTRODUCTION

In classical physics, matter and energy were treated as distinct entities—matter exhibited particle-like properties, while energy (e.g., light) displayed wave-like behavior. However, experiments in the 20th century revealed that both light and matter can exhibit *both* wave-like and particle-like properties depending on the context. This fundamental concept, known as **wave-particle duality**, forms the bedrock of modern quantum physics.

CONCEPT OF WAVE-PARTICLE DUALITY AND HISTORICAL CONTEXT

The **wave-particle duality** refers to the quantum mechanical concept that every particle or quantum entity can exhibit both wave and particle characteristics. This idea was first introduced in the early 20th century through two critical observations:

- **Photoelectric Effect (1905):** Albert Einstein explained how light, traditionally considered a wave, could behave like a stream of particles (photons) when it knocks electrons off a metal surface. This demonstrated the **particle nature of light**.
- **Electron Diffraction (1927):** Electrons, usually treated as particles, were found to produce interference patterns when passed through a crystal lattice or a double-slit apparatus—behaviors typical of waves. This demonstrated the **wave nature of matter**.

These discoveries suggested that both **light** and **particles of matter (like electrons, protons, etc.)** can behave as both **waves and particles**, challenging the classical distinction between the two.

DE BROGLIE'S HYPOTHESIS AND DE BROGLIE WAVELENGTH

In 1924, **Louis de Broglie**, a French physicist, extended the wave-particle duality of light to **material particles** such as electrons, protons, neutrons, and even larger objects theoretically. He proposed that every moving particle has a **wave associated with it**, called the **matter wave**.

De Broglie's Equation:

$$\lambda = \frac{h}{mv}$$

Where:

λ = wavelength of the matter wave (in meters)

h = Planck's constant (6.626×10^{-34} Js)

m = mass of the particle (in kg)

v = velocity of the particle (in m/s)

This equation suggests that lighter particles moving at higher velocities exhibit more pronounced wave-like behavior. For macroscopic objects, the wavelength is so small it is undetectable, but for microscopic particles like electrons, the wave nature becomes significant and observable.

EXPERIMENTAL EVIDENCE FOR WAVE NATURE OF PARTICLES

One of the most compelling pieces of evidence for the wave-like nature of particles came from electron diffraction experiments.

Davisson-Germer Experiment (1927):

- Conducted by Clinton Davisson and Lester Germer.
- Electrons were directed at a nickel crystal.
- The scattered electrons formed a diffraction pattern on a detector—something only waves do.
- This confirmed that electrons exhibit wave-like behavior, validating de Broglie's hypothesis.

Electron Double-Slit Experiment:

- When electrons are passed through a double slit one at a time, they still produce an interference pattern on the screen.
- This shows that even individual particles exhibit **self-interference**, a hallmark of wave behavior.

These experiments demonstrate that particles such as electrons, under suitable conditions, behave not just like particles but also like waves.

IMPLICATIONS OF WAVE-PARTICLE DUALITY

1. Limits of Classical Physics:

The duality showed that classical physics couldn't fully describe the behavior of subatomic particles. It led to the development of **quantum mechanics**, a new framework to explain such phenomena.

2. Probability in Quantum Physics:

Since particles behave like waves, their positions and momenta cannot be precisely predicted. Instead, they are described in terms of **probability waves** (wave functions).

3. Uncertainty Principle:

Werner Heisenberg's uncertainty principle emerges naturally from wave-particle duality. The more precisely we know a particle's momentum, the less precisely we can know its position, and vice versa.

4. Atomic Structure:

The wave nature of electrons explains why electrons exist in discrete energy levels around the nucleus and not spiral into it, as classical physics would suggest.

TECHNOLOGIES AND APPLICATIONS BASED ON WAVE-PARTICLE DUALITY

1. Electron Microscope:

Utilizes the wave properties of electrons to resolve structures much smaller than what light microscopes can detect. Since electrons have much smaller wavelengths than visible light, they offer higher resolution.

2. **Scanning Tunneling Microscope (STM):**

Relies on quantum tunneling of electrons, a direct result of wave-like behavior, to image surfaces at the atomic level.

3. **Quantum Mechanics & Quantum Computing:**

The entire field of **quantum physics** is built upon the principles of wave-particle duality. Quantum computers leverage the dual nature of particles (qubits) to perform complex computations more efficiently than classical computers.

4. **Particle Accelerators:**

Use wave-particle properties to accelerate subatomic particles to high energies and study their interactions.

5. **Semiconductor Physics:**

The behavior of electrons in semiconductors, which power all modern electronics, is described using wave mechanics.

EVALUATION QUESTIONS (WAEC/NECO PAST QUESTIONS)

1. *(WAEC June 2015 Q2)*

(a) Define wave-particle duality.

(b) State two experimental observations supporting the wave nature of electrons.

2. *(WAEC June 2022 Q1)*

A particle of mass $9.1 \times 10^{-31} \text{ kg}$ moves with a speed of $2.0 \times 10^6 \text{ ms}^{-1}$. Calculate its de Broglie wavelength.

(Take Planck's constant $h = 6.63 \times 10^{-34} \text{ Js}$)

$h = 6.63 \times 10^{-34} \text{ Js}$

3. *(NECO 2020 Q3)*

Explain why the wave nature of a moving car is not observable, but that of an electron is.

4. *(WAEC June 2017 Q4)*

Describe the significance of de Broglie's hypothesis in the development of modern atomic theory.

ASSIGNMENT

1. **Derive** the de Broglie equation from the relationship between momentum and wavelength.
2. **State and explain** two technological devices that utilize the wave nature of particles.
3. **Research and write** a short report on how quantum mechanics is applied in modern computing (Quantum Computing).
4. *(WAEC June 2021)*

A proton has a mass of $1.67 \times 10^{-27} \text{ kg}$ and is moving with a velocity of $5.0 \times 10^5 \text{ ms}^{-1}$. Calculate its de Broglie wavelength.

(Use $h = 6.63 \times 10^{-34} \text{ Js}$)



WEEK 10 - ENERGY AND SOCIETY

MEANING AND FORMS OF ENERGY

Definition of Energy

Energy is the capacity to do work or cause change. It is a fundamental physical quantity and is required in all activities of life, from motion and growth to production and communication.

Energy exists in various forms and can be transformed from one form to another.

FORMS OF ENERGY

1. **Kinetic Energy:** Energy possessed by a body due to its motion.
 $KE = \frac{1}{2}mv^2$ where m is mass and v is velocity.
2. **Potential Energy:** Energy stored in a body due to its position or configuration.
 $PE = mgh$
where m is mass, g is acceleration due to gravity, and h is height.
3. **Thermal Energy:** The internal energy of an object due to the random motion of its particles. It is related to temperature.
4. **Electrical Energy:** Energy caused by the movement of electric charges (electrons). It powers electrical devices and systems.
5. **Chemical Energy:** Energy stored in chemical bonds of substances. It is released during chemical reactions, such as combustion or metabolism.
6. **Nuclear Energy:** Energy stored in the nucleus of atoms, released during nuclear fission or fusion.
7. **Sound Energy:** Energy carried by sound waves, resulting from vibrations of particles.
8. **Radiant or Light Energy:** Energy carried by electromagnetic waves, especially visible light, X-rays, UV rays, etc.

Major Sources of Energy

Energy sources are categorized as:

1. Renewable Sources – Naturally replenished on a human timescale:

- **Solar energy** – from the sun.
- **Wind energy** – from moving air.

- **Hydropower** – from moving water.
- **Biomass** – from organic matter.
- **Geothermal energy** – from Earth's internal heat.
- **Tidal energy** – from ocean tides.

2. Non-Renewable Sources – Finite and depletable:

- **Fossil fuels** – coal, petroleum, natural gas.
- **Nuclear fuels** – uranium and plutonium.

ENERGY USE AND ITS IMPACT ON SOCIETY

1. Social Development:

- Energy powers transportation, lighting, cooking, and communication.
- It supports education, healthcare, and water supply systems.
- Availability of energy enhances quality of life and economic opportunities.

2. Technological Advancement:

- Energy is central to the development of industries, manufacturing, and digital systems.
- Electrical energy supports AI, robotics, and internet technologies.

3. Health:

- Access to energy ensures clean water, storage of medicines, operation of hospitals.
- However, pollution from fossil fuels can lead to respiratory diseases and cancer.

4. Environment:

- Fossil fuel use leads to air pollution, global warming, and climate change.
- Deforestation for biomass leads to habitat loss and reduced biodiversity.
- Oil spills, acid rain, and radioactive waste also harm the ecosystem.

ENERGY CONSERVATION AND SUSTAINABILITY

Energy Conservation means using energy more efficiently and reducing wastage.

Methods Include:

- Using energy-saving appliances (LEDs, efficient refrigerators).
- Switching off unused devices.
- Insulating buildings to reduce heating/cooling needs.
- Adopting public transport and reducing reliance on personal cars.

Importance of Sustainable Energy Use:

- Reduces environmental degradation.
- Ensures availability of energy for future generations.
- Promotes economic resilience and reduces energy costs.
- Aligns with global climate goals like the Paris Agreement.

Evaluation of Different Energy Sources

Source	Advantages	Limitations
Solar	Clean, renewable, low maintenance	Weather dependent, high setup cost
Hydro	Renewable, reliable, efficient	Ecosystem disruption, expensive
Wind	Renewable, zero emissions	Noise, aesthetic impact, variable output
Fossil Fuels	High energy output, easy transport	Pollution, finite, global warming
Nuclear	Large energy yield, low emissions	Radioactive waste, risk of accidents
Biomass	Reduces waste, renewable	Air pollution, deforestation risks

Step 7: Global and National Energy Challenges

- 1. Overdependence on Fossil Fuels:**
 - Leads to depletion and pollution.
- 2. Energy Insecurity:**
 - Many regions lack access to reliable electricity (especially in rural Africa).
- 3. Climate Change:**
 - Resulting from greenhouse gas emissions tied to energy use.
- 4. Geopolitical Conflicts:**
 - Energy resources can trigger tensions and wars.
- 5. High Cost of Renewable Transition:**

- Initial capital for solar, wind, and hydro projects is high.

Practical Solutions

- **Investment in Renewable Energy:** Government and private sector support for solar, wind, and hydro projects.
- **Public Awareness:** Education campaigns on energy efficiency.
- **Policy Reforms:** Enforce energy standards and encourage green innovation.
- **Technology Transfer:** Adopt clean technologies from developed nations.
- **Diversification:** Mix of energy sources to ensure reliability and reduce pressure on any single source.

Evaluation (WAEC/NECO Past Questions)

1. **(WAEC 2019 Q4):** State two advantages and two disadvantages of nuclear energy.
2. **(NECO 2020 Q5):** Define energy and list four different forms of energy.
3. **(WAEC 2017 Q2):** Distinguish between renewable and non-renewable sources of energy with two examples each.
4. **(WAEC 2018 Q8):** Mention three ways by which energy can be conserved in homes and industries.

Assignment

1. Discuss three ways in which energy use affects the environment.
2. With suitable examples, compare the advantages and disadvantages of hydroelectric power and fossil fuels.
3. Suggest two realistic ways Nigeria can reduce its reliance on fossil fuels.
4. Identify five everyday practices that promote energy conservation.